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Delayed Hydrogen Exsolution in Sub-Neptune Exoplanets under Magma-Envelope Coupling

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of F. Laga & S. Vanhulle (2025)

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All data, code and figures are reproducible from the directory `research/spinodal_onset` of the Atmodeller fork on the branch `feat/spinodal-metastability-study`. Source data of Gilmore & Stixrude (2026) are used under the Creative Commons Attribution 4.0 licence.

Chapter 2 reuses wording from the bachelor thesis of F. Laga and S. Vanhulle, *Atmospheric Composition of Sub-Neptune Exoplanets under Magma-Envelope Coupling* (KU Leuven, 2025).

Preface

We would like to thank Dr. Kaustubh Hakim for his guidance and teachings during the bachelor thesis on which this work builds. We would also like to thank Siemen Vanhulle for the joint thesis work, from which parts of the text are reused.

We acknowledge the use of AI tools for writing code, running the analysis and drafting this text. Every result, interpretation and conclusion must be checked by the authors before submission.

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Abstract

Young sub-Neptunes may hold magma that is fully miscible with their hydrogen envelopes. On cooling, hydrogen exsolves and releases heat that none of the evolution models we checked includes. We add energy, nucleation and cooling-path functions to the open-source code Atmodeller and apply them to the 2026 H_2–$MgSiO_3$ mixing model and its source simulations.

At 1–3 wt% hydrogen and 2–5 GPa, exsolution releases 35–93 MJ per kg of hydrogen in refits to the simulated excess enthalpies, and 51–64 MJ kg^{-1} in the published fit, which underestimates the simulated excess enthalpy of the most hydrogen-poor mixture (24.5 against 36.3 ± 5.9 kJ mol^{-1}). A convecting interior degasses into the envelope without nucleation. At 3 wt% and 2 GPa, its exsolution heat over the first 300 K below the binodal is 2.3 (1.9–2.9) times the sensible heat, and its effective heat capacity rises up to 7.0-fold if the hydrogen leaves as pure H_2. Scaled onto a published evolution built on an earlier phase diagram, the heat is 4–7% of the energy radiated.

Away from the envelope, homogeneous nucleation with the simulated interface width ends metastability 1.2–2.9 K below the binodal (refit: 0.2–1.6 K) and withholds at most 1.3% of the hydrogen. The 414 K binodal–spinodal window of the published fit is thus an unreached mean-field bound. Refits give median windows of 41–242 K. A 10% curvature measurement at 3600 K would lower the standard deviation of the spinodal only from 80 to 64 K, and only through the assumed model forms.

Samenvatting

Jonge sub-Neptunussen kunnen magma bevatten dat volledig mengbaar is met hun waterstofomhulsel. Bij afkoeling ontmengt waterstof, en de vrijgekomen warmte ontbreekt in alle evolutiemodellen die wij nagingen. Wij voegen functies voor energie, kiemvorming en afkoelingspaden toe aan de opensourcecode Atmodeller en passen ze toe op het mengmodel voor H_2–$MgSiO_3$ uit 2026 en op de simulaties waarop het berust.

Bij 1–3 massa% waterstof en 2–5 GPa maakt ontmenging 35–93 MJ per kg waterstof vrij in nieuwe fits aan de gesimuleerde exces-enthalpieën, en 51–64 MJ kg^{-1} in de gepubliceerde fit, die de gesimuleerde exces-enthalpie van het waterstofarmste mengsel onderschat (24.5 tegenover 36.3 ± 5.9 kJ mol^{-1}). Een convectief binnenste ontgast zonder kiemvorming in het omhulsel. Bij 3 massa% en 2 GPa is zijn ontmengingswarmte over de eerste 300 K onder de binodaal 2.3 (1.9–2.9) keer de voelbare warmte, en stijgt zijn effectieve warmtecapaciteit tot het 7.0-voudige als de waterstof als zuiver H_2 vertrekt. Geschaald op een gepubliceerde evolutie met een ouder fasediagram is de warmte 4–7% van de uitgestraalde energie.

Ver van het omhulsel beëindigt homogene kiemvorming met de gesimuleerde grensvlakbreedte de metastabiliteit 1.2–2.9 K onder de binodaal (nieuwe fit: 0.2–1.6 K), en blijft hoogstens 1.3% van de waterstof achter. Het binodaal-spinodaalvenster van 414 K in de gepubliceerde fit is dus een onbereikte gemiddeldveldbovengrens. Nieuwe fits geven mediane vensters van 41–242 K. Een krommingsmeting met 10% precisie bij 3600 K zou de standaardafwijking van de spinodaal slechts van 80 naar 64 K verkleinen, en alleen via de aangenomen modelvormen.

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List of Abbreviations and Symbols

Abbreviations

DFT-MD	Density-functional-theory molecular dynamics
ED Table	Extended Data Table of Gilmore & Stixrude (2026)
EOS	Equation of state
MAP	Maximum a posteriori (best-fitting) parameters
RK	Redlich–Kister composition series
xLMA	Extended law of mass action

Symbols

x	H_2 mole fraction on the oxide basis, Eq. (2.6)
y	H_2 mole fraction in native ($MgSiO_3$, H_2) formula units
w	Local hydrogen mass fraction of a closed parcel
g_{mix}	Molar Gibbs energy of mixing per mole of mixing units
$W(x)$	Excess enthalpy at zero pressure, Eq. (2.5)
A, B	Interaction parameters of the published fit
τ, π	Temperature and pressure scales of the excess energy
H_{ex}, S_{ex}, V_{ex}	Excess enthalpy, entropy and volume
T_b	Binodal temperature: first equilibrium separation on cooling
T_{sp}	Spinodal temperature: loss of local stability
ΔT_{ms}	Metastable interval $T_b - T_{\mathrm{sp}}$
f_{H}	Fraction of hydrogen in the H-rich phase at T_{sp}
T_C, x_C	Critical temperature and composition
σ_x, σ_T	Composition and phase-state uncertainties
χ^2	Misfit between model and source data, Eq. (2.11)
R	Gas constant
$R_\oplus$, $M_\oplus$	Earth radius and Earth mass

1. Introduction

1.1 Miscible Interiors of Sub-Neptunes

Planets between Earth and Neptune in size are common. Among close-in Kepler planets, a gap in the radius distribution at 1.5–2.0 $R_\oplus$ separates smaller super-Earths from larger sub-Neptunes [11]. The larger planets are thought to be rocky interiors with envelopes of low-density gas [11]. In the evolution models discussed here, that gas is hydrogen, and a fiducial planet of six Earth masses holds 3% of its mass as hydrogen [30].

Gilmore & Stixrude [13] showed with density-functional-theory molecular dynamics (DFT-MD) that hydrogen and liquid $MgSiO_3$ are completely miscible above a critical temperature T_C. In their fit, T_C falls from about 3,500 K at 2 GPa to 2,600 K at 10 GPa. They found such conditions at the core–envelope boundary over a wide range of plausible pressures and temperatures. Young et al. [39] argued that full miscibility is the common state of sub-Neptune interiors. The interior and the envelope then form one fluid, not a melt in contact with a gas. Evolution models now include this miscibility. In the models of Rogers et al. [30] and Rogers & Schlichting [29], the binodal moves into the interior as the planet cools. Hydrogen leaves the melt for the envelope as this boundary of the miscibility gap passes. Young & Werlen [38] and Tang et al. [34] also couple dissolved hydrogen to the evolution of the planet. Tang et al. treat its release as effectively instantaneous. These models track where the hydrogen goes. Their energy equations do not include the heat that its separation releases (Section 3.9). Tejada Arevalo et al. [35] study the other side of the gap: silicate that rains out of the envelope, above a mantle that exchanges no hydrogen with it.

1.2 Heat of Exsolution

Mixing hydrogen into silicate melt absorbs heat. The excess enthalpy of the source simulations is positive at every simulated composition [14], so separation releases heat. The process resembles crystallization, which releases latent heat. Sub-Neptune evolution models count the latent heat of a solidifying mantle [34, 35]. A closer precedent is helium rain in Jupiter and Saturn, where helium demixes from metallic hydrogen. Evolution models of these planets count the energy released as the helium sinks, mainly as gravitational energy [10, 23, 32]. The latent heat of helium demixing is estimated at about 200 MJ per kg of helium. Because the droplets redissolve deeper, it moves heat within the planet but adds none [26]. A miscible sub-Neptune interior loses its hydrogen to the envelope for good, so the heat is a net source, and its evolution models have no term for it. How large this heat is, and how it changes the cooling of the interior, are the main questions of this work.

1.3 Metastability and Delayed Exsolution

Equilibrium models let hydrogen exsolve as soon as the composition enters the miscibility gap, at the binodal [29, 30]. A homogeneous mixture just inside the binodal is *metastable*, however. Forming the new phase requires nucleation, which must overcome a free-energy barrier [9, 17]. Only at the **spinodal** does the mixture become unstable to small fluctuations and decompose spontaneously [5, 6]. There, the curvature of the Gibbs energy with respect to composition vanishes. Between the two lies a metastable window (Figure 1.1).

Young et al. [39] already plotted the binodal, the spinodal and this window for H_2–$MgSiO_3$ (their Fig. 2), using the 2025 preprint fit [33]. They did not ask how long the window delays separation, or whether nucleation lets a cooling parcel reach the spinodal. In silicate magmas, delayed degassing through suppressed nucleation is well documented [22], and nucleation on crystals and other interfaces may dominate [31]. Nishiwaki [27] derived a spinodal for hydrous magma from solubility data through an assumed solution model.

Such an inference has a known weakness. The spinodal is a property of the homogeneous free energy inside the gap, and simulations of coexisting phases do not sample that state. A spinodal derived from a fitted mixing model therefore inherits the assumed functional form. We test how far the existing data constrain it.

1.4 Contributions and Scope

We study binary H_2–$MgSiO_3$ mixtures with the 2026 mixing model [13] and its source simulations [14]. We cover local hydrogen mass fractions of 0.1–20%, pressures of 2–10 GPa and temperatures of 2,000–4,000 K. The local mass fractions describe the melt at the top of the interior, not the hydrogen budget of the whole planet. This work makes four contributions.

1. **Open-source tools.** We add energy, nucleation and cooling-path functions to the open-

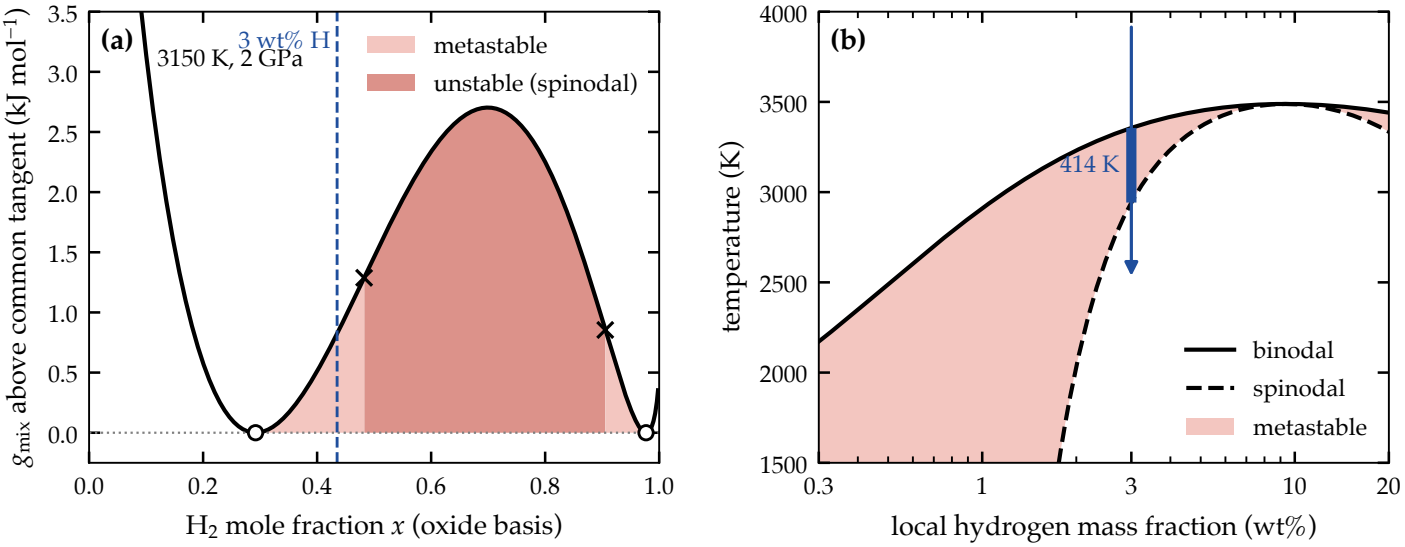


Figure 1.1: ***Binodal, spinodal and the metastable window of a cooling H_2–$MgSiO_3$ parcel.*** *(a) Homogeneous Gibbs energy of mixing at 2 GPa and 3150 K above its common tangent, for the published mixing model [13]. Circles mark the coexisting compositions (binodal), crosses the inflection points (spinodal). A parcel with 3 wt% local hydrogen (dashed line) is metastable here. (b) Binodal and spinodal temperatures at 2 GPa as a function of local hydrogen mass fraction. A 3 wt% parcel cooling at fixed pressure (arrow) enters the miscibility gap at the binodal but becomes unstable only at the spinodal.*

source code Atmodeller [3], with a composition-consistent basis for the 2026 model. A separate scalar code reproduces the Atmodeller binodals and spinodals to within 10^{-6} K. The full Atmodeller equilibrium solver reproduces the mixing enthalpy along one published-fit parcel path (3 wt% at 2 GPa, separating at the spinodal) to a relative difference of 9×10^{-16} over 80 solves (Section 2.6 and Appendix D).

2. **The heat of exsolution.** We compute the heat released per kilogram of exsolved hydrogen from the same free energy that sets the binodal. We test it against the simulated excess enthalpies and follow its effect on a convecting interior (Sections 3.6 and 3.9).
3. **Near-equilibrium kinetics.** We compute the homogeneous nucleation barrier in square-gradient theory. It ends metastability 1.2–2.9 K below the binodal for the published fit (Section 3.7).
4. **What the source data constrain.** We audit the published fit against its source simulations and refit four model classes. The data constrain the heat, but they do not fix the spinodal (Sections 3.3–3.5).

We do not run a planetary evolution model. The convecting interior is well mixed, and its composition follows the hydrogen-poor binodal at its top (Section 3.9). The planet-wide estimate post-processes a published energy budget [30]. The mixture contains no helium or iron. Like the source fit, all model classes take the excess enthalpy, entropy and volume as independent of temperature.

2. Methodology

This section outlines the complete modeling framework used to determine the equilibrium and stability limits of hydrogen exsolution at the magma-atmosphere boundary of sub-Neptune exoplanets, called the core–envelope boundary in [13]. The implementation is performed using Atmodeller [3], an open-source Python library designed to resolve coupled chemical equilibrium and mass-balance constraints in gas-melt environments. In the following sections, we outline the conceptual basis, numerical approach, chemical system, and software validation.

2.1 Formulation and Numerical Solution

2.1.1 *Solver Architecture*

Users define the gaseous species, from which the software generates an independent reaction basis using Gaussian elimination (Leal et al. 2016 [21]). At the interface, the solver determines chemical equilibrium among gas species and any specified dissolved species within the melt, subject to elemental mass conservation. The solver determines the equilibrium state by minimizing the total Gibbs free energy using the optimistix framework, with

$$G = \sum_i n_i \mu_i(T, P), \tag{2.1}$$

where n_i is the number of moles and μ_i is the chemical potential of species i, which includes EOS corrections and melt interactions. Atmodeller employs the extended law of mass action (xLMA; Leal et al. 2016 [21]) to build a closed nonlinear system that integrates equilibrium chemistry with elemental mass balance, structured as a root-finding problem, and leverages JAX for high-performance computations (Bradbury et al. 2018 [4]). The nonlinear system is resolved using robust Newton–type root–finders augmented with line–search and trust–region safeguards.

For this work we extended the phase-equilibrium module of Atmodeller. A binary mixing model now supplies the nonideal activities of two condensed components, here $MgSiO_3$ and H_2. The same Gibbs energy yields the homogeneous activities, the coexisting compositions, the local stability and the equilibrium (convex-envelope) free energy, so that all quantities in this work follow from one function.

2.1.2 *Phase Equilibrium and Local Stability*

For a binary mixture with molar Gibbs energy of mixing $g_{\mathrm{mix}}(x)$, two phases α and β coexist when both component chemical potentials are equal,

$$\mu^{\alpha}_{\mathrm{sil}} = \mu^{\beta}_{\mathrm{sil}}, \qquad \mu^{\alpha}_{\mathrm{H_2}} = \mu^{\beta}_{\mathrm{H_2}}, \tag{2.2}$$

which is the common-tangent construction. The locus of coexisting compositions as a function of temperature is the **binodal**. A homogeneous mixture is locally stable as long as

$$\frac{\partial^2 g_{\mathrm{mix}}}{\partial x^2} > 0, \tag{2.3}$$

and the locus where this curvature vanishes is the **spinodal**. Between the binodal and the spinodal a homogeneous mixture is metastable: it is stable against small composition fluctuations, but a nucleus of the other phase that exceeds a critical size lowers the free energy [9]. Inside the spinodal it decomposes spontaneously [5]. Both conditions are evaluated with bracketed root finding on the two stable branches, which excludes the trivial equal-composition solution, and with implicit derivatives for use inside the Atmodeller solver.

2.2 Hydrogen–Silicate Mixing Model

Gilmore & Stixrude [13] describe their phase-coexistence and homogeneous DFT-MD simulations with an asymmetric regular solution model. We write their equation 8 as

$$\begin{aligned} g_{\mathrm{mix}} = W(x)\left(1 - \frac{T}{\tau} + \frac{P}{\pi}\right) \\ + RT\left[x\ln x + (1-x)\ln(1-x)\right], \end{aligned} \tag{2.4}$$

with the excess enthalpy at zero pressure

$$W(x) = x(1-x)\left[A(1-x) + Bx\right], \tag{2.5}$$

excess entropy W/τ and excess volume W/π. The published parameters are $A = 75\,\mathrm{kJ\,mol^{-1}}$, $B = 424\,\mathrm{kJ\,mol^{-1}}$, $\tau = 4{,}400\,\mathrm{K}$ and $\pi = -35\,\mathrm{GPa}$. The composition variable is the molar fraction of the H_2 component on an oxide basis,

$$\begin{aligned} x &= \frac{N_{\mathrm{H_2}}}{N_{\mathrm{H_2}} + N_{\mathrm{MgO}} + N_{\mathrm{SiO_2}}} \\ &= \frac{N_{\mathrm{H}}/2}{N_{\mathrm{H}}/2 + 2(N_{\mathrm{Mg}} + N_{\mathrm{Si}} + N_{\mathrm{O}})/5}. \end{aligned} \tag{2.6}$$

The paper does not state which component carries A and which carries B. The critical temperature is the same for both choices, but the shape of the miscibility gap is not. Assigning B to the H_2-rich side (Eq. 2.5) reproduces the published curves of their Figure 2 and their statement that the hydrogen-poor phase holds more hydrogen than the hydrogen-rich phase holds silicate (Section 2.6).

Composition basis. Atmodeller uses native formula units, with the H_2 fraction $y = n_{\mathrm{H_2}}/(n_{\mathrm{H_2}} + n_{\mathrm{MgSiO_3}})$. One $MgSiO_3$ unit contains two oxide units, so

$$x = \frac{y}{2-y}, \qquad G = \left(2\,n_{\mathrm{MgSiO_3}} + n_{\mathrm{H_2}}\right) g_{\mathrm{mix}}(x). \tag{2.7}$$

The second relation keeps the free energy extensive; the native chemical potentials are $\mu_{\mathrm{MgSiO_3}} = 2\mu_{\mathrm{oxide}}$ and $\mu_{\mathrm{H_2}}$. Converting only the endpoint compositions is not sufficient, because the native curvature, activities and phase amounts all depend on the factor $2 - y$. We implemented this conversion as a static multiplicity of the silicate component, available through the constructor `hydrogen_silicate_gilmore26` of the class `SubregularSolution`.

Controls. As controls we retain the historical 2025 parameterization of Rogers, Young & Schlichting [30] on its own formula-unit basis, and a deliberately incorrect import of the 2026 parameters without the basis conversion. The latter is a diagnostic, never a recommended model.

2.3 Mean-Field Bound on Delayed Exsolution

We follow a closed, initially homogeneous parcel that cools at fixed pressure. Its local hydrogen mass fraction w fixes y and hence x. On cooling it first enters the miscibility gap at the binodal temperature $T_b(w, P)$ and later becomes locally unstable at the spinodal temperature $T_{sp}(w, P)$. Because Eq. (2.4) is linear in T, the spinodal temperature follows analytically from Eq. (2.3):

$$T_{sp} = \frac{-W''(x)\,(1 + P/\pi)}{R/[x(1-x)] - W''(x)/\tau}. \tag{2.8}$$

We compare two limiting histories. In the **equilibrium limit** the parcel separates as soon as it crosses T_b. In the **delayed limit** nucleation is suppressed until T_{sp}, after which the parcel equilibrates rapidly. We report the metastable interval

$$\Delta T_{ms} = T_b - T_{sp}, \tag{2.9}$$

and the fraction of all hydrogen that the equilibrium limit already places in the H-rich phase at T_{sp},

$$f_H = \frac{\phi\, y_{rich}}{y_0}, \qquad \phi = \frac{y_0 - y_{poor}}{y_{rich} - y_{poor}}, \tag{2.10}$$

where ϕ is the native mole fraction of the H-rich phase from the lever rule. The quantity f_H is the largest inventory that the delayed limit can withhold from separation. It is a bound between two assumptions, not a computed release rate: both histories conserve all hydrogen, and the delayed parcel still holds its hydrogen dissolved in the silicate. The delayed limit is a mean-field upper bound. Section 3.7 shows that, with the simulated interface width, nucleation ends metastability 1.2–2.9 K below the binodal.

2.4 Model Configurations and Parameter Space

The results presented in the Results are derived from sweeps over four parameters. When varying a specific parameter, the other three are held at their baseline values unless otherwise noted.

1. **Local Hydrogen Mass Fraction (0.1% – 20%):** This range spans parcels from silicate-rich melt to hydrogen-rich fluid at the magma-atmosphere boundary.
   Baseline: $w = 3$ wt%, a composition that crosses both the binodal and the spinodal above 2,000 K.
2. **Pressure (2 – 10 GPa):** This range covers most of the magma-atmosphere boundary pressures of the sub-Neptune models compiled in [13].
   Baseline: $P = 2$ GPa, the pressure of the best-sampled simulations.
3. **Temperature (2,000 K – 4,000 K):** This window covers the fluid regime of the simulations. We report spinodals below 2,000 K separately and test stopping temperatures of 2,500 and 3,000 K, because neither is a computed melting boundary.
4. **Mixing Model:** the published 2026 fit (baseline), the historical 2025 fit and the incorrect-basis control, and the recalibrated model classes of Section 2.5.

In total the onset grid contains 20,664 rows and the cooling paths 10,827 states. A further 375 deterministic stress cases change A, B, τ and $|\pi|$ by $\pm 10\%$, individually and in combination. These stress cases are not calibrated uncertainties; Section 2.5 replaces them with data-driven posteriors.

2.5 Source-Data Audit and Recalibration

2.5.1 *Source Data*

The source simulations are public [14]. Extended Data (ED) Table 1 lists 14 phase-coexistence runs: four that kept two phases, with the composition of each phase, and ten that became one homogeneous phase, with their bulk composition. The quoted composition uncertainty is ± 0.05 in x. ED Table 2 lists homogeneous simulations at 5 GPa and 4,000 K: three mixtures and the two endmembers, with enthalpy, entropy and volume per atom. We reconstruct excess properties per mole of mixing units (Eq. 2.6) and remove the ideal mixing entropy from the total entropy. The endmember errors are shared between the three mixtures, so we propagate a full covariance matrix. The endmembers were simulated at 5.00 GPa and the mixtures at 5.06–5.08 GPa. This offset biases the excess properties with a known sign, so we correct their means and do not treat it as noise. We move the endmembers to the mixture pressure with $\Delta H = V\Delta P$, which neglects thermal expansion and so is an upper bound, and $\Delta V = -V\Delta P/K$. The source gives no bulk moduli; we assume $K = 15$ GPa for the hydrogen fluid and 30 GPa for the silicate, applied to the endmember volumes. The correction lowers the excess enthalpies by 0.8–1.4 kJ mol^{-1} and raises the excess volumes by 0.05–0.06 cm^3 mol^{-1}. With $K = 20$ GPa for both endmembers, the best-fit interval of class M1 changes by less than 1 K. Finally, we digitized the published Figure 2a, which shows the same coexistence runs, from the CC-BY figure file.

Table 2.1: ***Model classes used in the recalibration.*** *All classes have constant excess enthalpy, entropy and volume coefficients. The last column counts free parameters.*

Class	Excess Gibbs energy	n
M1	published form, Eq. (2.4)	4
M2	subregular H, S and V, independent	6
M3	RK order 2, published T, P scaling	5
M4	RK order 3, published T, P scaling	6

2.5.2 *Likelihood and Model Classes*

We compare models with the data through three terms,

$$\chi^2 = \chi^2_{\text{phase}} + \chi^2_{\text{comp}} + \chi^2_{\text{hom}}. \tag{2.11}$$

The first term uses the most robust information in the source: whether each of the 14 runs kept one or two phases. A model that misplaces a run pays $[(T_{\text{run}} - T_b)/\sigma_T]^2$, with T_b the model binodal temperature at the run's bulk composition and $\sigma_T^2 = (50\,\text{K})^2 + (100\,\text{K}\,\text{GPa}^{-1}\,\sigma_P)^2$, which adds the quoted pressure error. The second term compares the eight coexisting compositions, with the quoted ± 0.05 treated as a Gaussian standard deviation. If a model has no gap at a two-phase run, we place both predicted compositions at its least stable composition, the minimum of $\partial^2 g/\partial x^2$ on a grid with steps of 0.005. At every best fit this happens for the run at 1.65 GPa and 3,500 K, and the choice matters. At the M1 best fit, placing both compositions at x_C instead shortens the interval by about 14 K. Placing them at the bulk composition lengthens it by about 23 K. The third term compares the nine homogeneous excess properties with their covariance.

We fit four smooth model classes with constant excess enthalpy, entropy and volume, so that their spinodals follow in closed form. They are written as Redlich–Kister series [28],

$$G_{\text{ex}} = \sum_k (h_k - Ts_k + Pv_k)\, x(1-x)(1-2x)^k, \tag{2.12}$$

summarized in Table 2.1. Class M1 is the published form. M2 frees the shapes of the excess entropy and volume. M3 and M4 add one and two higher composition terms that share the published temperature and pressure scaling.

Priors are uniform within broad stated bounds; we sample $1/\tau$ and $1/\pi$ rather than τ and π. We draw posterior samples with an adaptive Metropolis algorithm [16]: eight chains of 12,000 steps, of which the first 3,000 adapt the proposal and are discarded. Each chain starts at the best fit plus a Gaussian offset with a standard deviation of 2% of each prior range, which is 1.5 to 4 posterior standard deviations for most parameters. We check convergence with the classic split-chain $\hat{R}$ statistic [12] of every parameter and derived quantity. We do not use the rank-normalized $\hat{R}$, and we compare our values with the threshold of 1.01 recommended for it [36]. All chains start near one best fit, so $\hat{R}$ cannot reveal a second mode far from it. Besides the full data set we fit five variants of class M1: with the Figure 2a compositions, with $\sigma_x = 0.15$ for the two runs closest to T_C, without the excess volumes, and with only the phase states plus either the compositions or the homogeneous data. For every retained sample we recompute T_b, T_{sp}, ΔT_{ms}, f_{H} and T_C at the baseline.

2.5.3 *Identifiability Test*

To test whether coexistence data can identify the spinodal at all, we add a nonnegative perturbation that lives only inside the miscibility gap,

$$\delta g = 64\,\lambda B_{\text{gap}}\, u^3(1-u)^3, \quad u = \frac{x - x_{\text{poor}}}{x_{\text{rich}} - x_{\text{poor}}}, \tag{2.13}$$

for $x_{\text{poor}} < x < x_{\text{rich}}$ and zero elsewhere. Here B_{gap} is the height of the published free energy above its common tangent at the middle of the gap. The perturbation and its first two derivatives vanish at the binodal, so all coexisting compositions, all equilibrium free energies and all homogeneous states outside the gap stay exactly the same.

At fixed composition, δg vanishes at the binodal and grows on cooling, so it adds a negative heat capacity $-T\,\partial^2 \delta g/\partial T^2$. Every nonnegative perturbation that vanishes outside the gap has this effect somewhere below the binodal. We therefore cap the amplitude so that every homogeneous state inside the gap keeps a positive heat capacity, at all compositions and between 2,000 K and T_C. The unperturbed model has no excess heat capacity, so its heat capacity is that of the endmembers. We take $3.2\,k_B$ per atom, the smallest value implied by the temperature dependence of the single-phase enthalpies in ED Table 1 (3.2–3.9 k_B per atom). The perturbation lowers the heat capacity most, by $14.8\,k_B$ per atom per unit λ, close to the critical point. This sets $\lambda \leq 0.22$. One amplitude holds for the whole free-energy surface: an amplitude chosen separately for each composition would leave negative heat capacities in metastable states of other compositions. The amplitudes $\lambda = 0$ to 0.22 are a counterexample to uniqueness, not a material model.

2.6 Verification of Thermodynamic Consistency

We verified the implementation in three independent ways. First, our implementation reproduces

the published model. Its critical temperatures are 3489 K at 2 GPa and 2643 K at 10 GPa, matching the published 3,500 and 2,600 K, and our binodals follow the digitized curves of the published Figure 2a with an RMS residual of at most 2.3 K (Figure 2.1a). The largest single deviation, 15 K, is about three pixels of the printed line.

Second, a separate scalar implementation, written without any code from Atmodeller, agrees with the Atmodeller binodal and spinodal temperatures to within 10^{-6} K over the grid (Figure 2.1b). This includes the binodal within 0.02 of x_C, where the coexisting phase lies close to the parcel and the two codes differ by at most 1×10^{-7} K. It locates spinodal compositions as the real roots of a polynomial and binodals by nested bracketed root finding. To decide whether a composition lies inside the gap, it compares the free energy with the tangent at that composition only on the other convex branches. So its errors are independent of the Atmodeller solver.

Third, the Atmodeller test suite passed 288 tests, including the extensive-energy derivative test of the basis conversion (Eq. 2.7), Gibbs–Duhem consistency and spinodal temperatures against independent roots. The study itself checks conservation of each native component in 27 native equilibrium solves and compares 30 spinodal temperatures with independent SciPy roots. The largest elemental and lever-rule errors are 2.2×10^{-16}.

2.7 Data Processing, Model Limitations and Assumptions

2.7.1 Data Processing

Model outputs were exported as CSV and JSON files with SHA-256 hashes of every artifact and analysis script. Data reduction was performed using pandas, and visualizations were generated using matplotlib. Key diagnostics reported in the Results include the binodal and spinodal temperatures, the metastable interval ΔT_{ms}, the withheld hydrogen fraction f_{H} and the critical temperature T_C.

2.7.2 Model Limitations and Assumptions

The current framework operates under several constraints. First, the mixture is a binary H_2–$MgSiO_3$ system; helium, iron and other volatiles are absent, and the source simulations used pure $MgSiO_3$. Second, all model classes have constant excess enthalpy, entropy and volume. The free energy is therefore linear in temperature and pressure, and the excess heat capacity is zero. Third, the parcel of Sections 2.3 and 3.8 is closed: it loses heat at fixed pressure but exchanges no mass. Its sensible heat comes from the end-member data without a pressure correction. Fourth, the convecting interior of Section 3.9 is uniform. Its melt follows the hydrogen-poor binodal at a fixed top pressure, and the exsolved hydrogen joins the envelope as pure H_2 (Appendix A.7). Its sensible heat follows the top temperature, and the model includes no gravitational, settling or envelope energy. Fifth, nucleation is homogeneous and follows square-gradient theory (Section 3.7 and Appendix A.8). One gradient coefficient, set by a simulated interface width, and one molar volume serve every composition, temperature and pressure. The nucleation rate enters only through the criterion $W^*/k_{\mathrm{B}}T = \ln(J_0 V t)$. Finally, growth after the first nucleus is assumed fast, so the parcel then takes its equilibrium state; diffusion-limited growth and spatial gradients within the reservoirs are not modelled.

3. Results

This chapter details the equilibrium, stability and energy of hydrogen exsolution from H_2–$MgSiO_3$ mixtures. We first characterize the binodal and spinodal of the published model. We then give the mean-field upper bound that the spinodal sets on delayed exsolution and on the hydrogen it could withhold. Next, the published fit is audited against its source simulations and recalibrated, and we test whether coexistence data can identify the spinodal. The chapter ends with the heat of exsolution, the nucleation limit on metastability, and the cooling of a closed parcel and of a convecting interior (Sections 3.6–3.9).

3.1 Phase Boundaries and the Metastable Window

Figure 3.1 shows the binodal and spinodal temperatures of the published model as a function of local hydrogen mass fraction at 2, 5 and 10 GPa. The binodal is flat-topped: its crest at 2 GPa lies at $T_C = 3489$ K and $x_C = 0.720$ ($w = 9.36$ wt%), and it falls only slowly away from x_C. The spinodal touches the binodal at the critical point and falls steeply on the hydrogen-poor side. Spinodals exist only above $w = 1.40$ wt%. Below this threshold a parcel can still separate in equilibrium, but its homogeneous state never becomes unstable, so separation always requires nucleation (Appendix A).

For the baseline parcel of 3 wt% hydrogen at 2 GPa, the binodal lies at 3356 K and the spinodal at 2942 K. The metastable interval is therefore 414 K (Table 3.1). It shrinks to 376 K at 5 GPa and 313 K at 10 GPa, because both onset temperatures scale with $1 + P/\pi$.

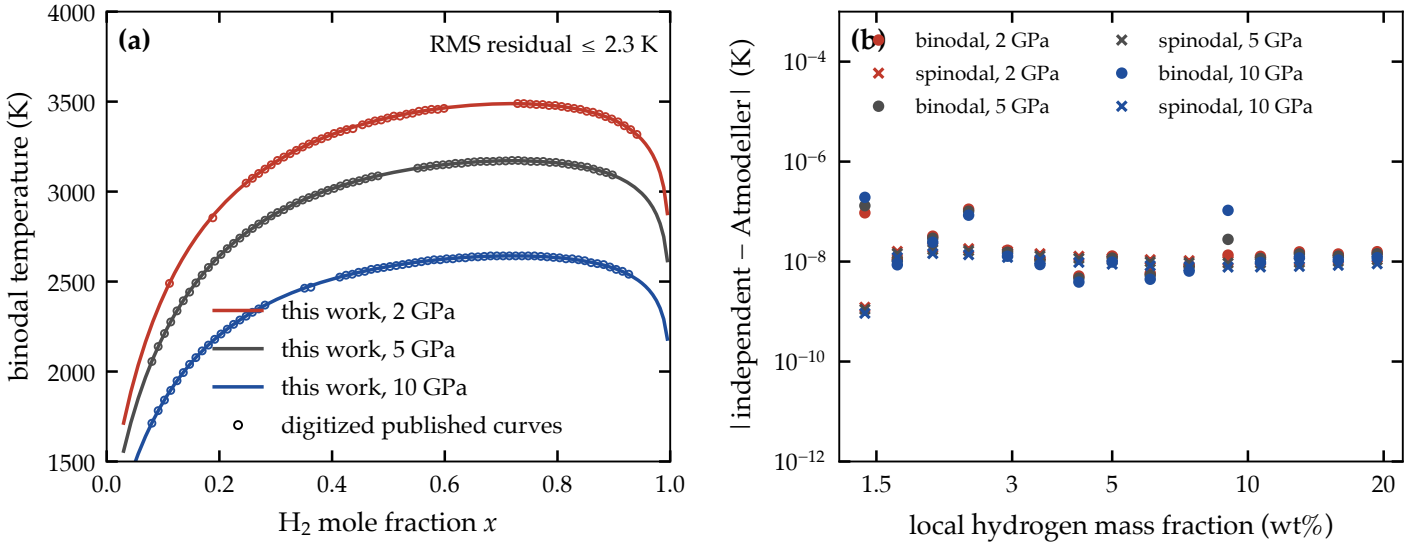


Figure 2.1: ***Internal consistency check of Atmodeller.*** *(a) Binodal temperatures of our implementation of Eq. (2.4) (lines) and the curves digitized from Figure 2a of [13] (circles) at 2, 5 and 10 GPa. (b) Absolute difference between a separate scalar implementation and Atmodeller for the binodal (circles) and spinodal (crosses) temperatures across the onset grid. The two codes share no solver code.*

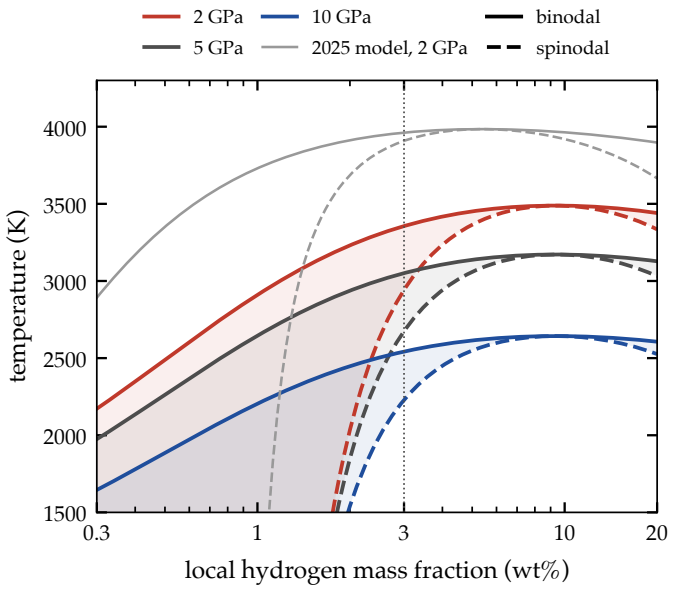


Figure 3.1: ***Binodal and spinodal temperatures of the published mixing model.*** *Solid lines show the binodal, dashed lines the spinodal, at 2, 5 and 10 GPa; shading marks the metastable window. Grey lines show the historical 2025 parameterization at 2 GPa. The dotted line marks the 3 wt% baseline.*

At 10 GPa the spinodal lies below the coldest two-phase simulation (2,500 K), so it is an extrapolation. The composition basis matters. The historical 2025 parameterization gives an interval of only 53 K, and importing the 2026 parameters without the basis conversion gives 46 K. With the correct conversion the interval is about nine times larger. Changing A, B, τ and $|\pi|$ by $\pm 10\%$ gives intervals of 329–504 K. These stress cases are not calibrated uncertainties; Section 3.4 derives uncertainties from the data.

Table 3.1: ***Onset temperatures for a parcel with 3 wt% local hydrogen.*** *f_H is the fraction of hydrogen that equilibrium places in the H-rich phase at the spinodal.*

Model, P	T_b (K)	T_{sp} (K)	ΔT_{ms} (K)	f_H (%)
2026 fit, 2 GPa	3356	2942	414	65.5
2026 fit, 5 GPa	3051	2675	376	65.5
2026 fit, 10 GPa	2542	2229	313	65.5
2025 fit, 2 GPa	3962	3908	53	39.0
Wrong basis, 2 GPa	3468	3422	46	36.1

3.2 Hydrogen Withheld in the Delayed Limit

Figure 3.2a follows the 3 wt% parcel as it cools. In the equilibrium limit the H-rich phase appears at T_b and grows on further cooling. In the delayed limit nothing separates until T_{sp}, where the parcel jumps to the equilibrium state. At the spinodal the equilibrium limit already places 65.5% of all hydrogen in the H-rich phase, while the delayed limit still holds it in the homogeneous parcel.

At 3,000 K and 2 GPa the baseline parcel is metastable. Equilibrium would place 61.4% of its hydrogen in the H-rich phase, which is 18.4 g of the 30 g of hydrogen in each kilogram of parcel, whereas the delayed limit has not separated at all. Equilibration would lower the Gibbs energy by 2.98 kJ per native mole. This is a free-energy difference, not a prediction of released heat.

The withheld fraction at the spinodal is the same at 2, 5 and 10 GPa to within 2×10^{-10}, or 7×10^{-6} within 0.1 wt% of the critical composition, where the lever rule becomes ill-conditioned (Figure 3.2b). It falls from 88% at 2 wt%, the lowest composition whose spinodal lies above 2,000 K, to about 21% near the critical composition. The pressure invariance

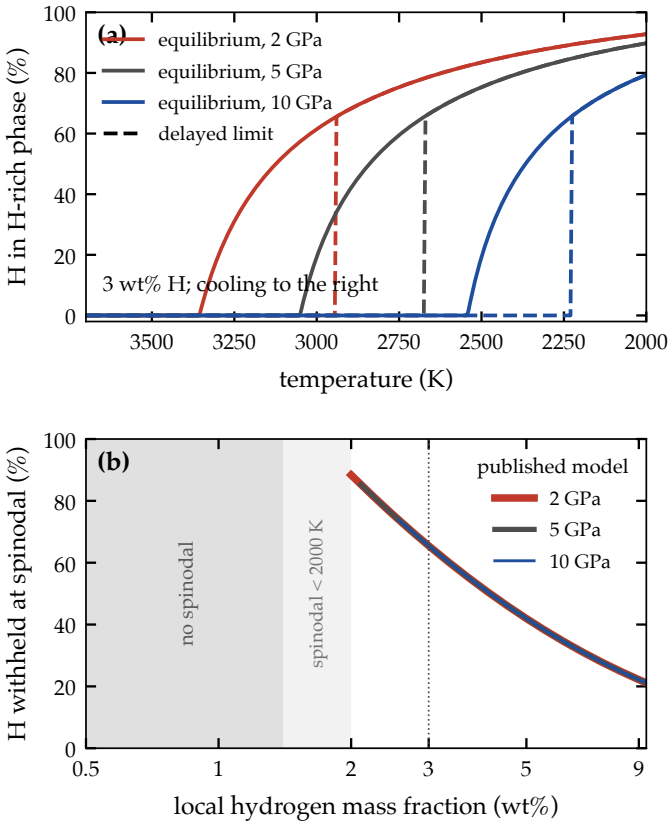


Figure 3.2: ***Hydrogen that the delayed limit withholds (mean-field bound).*** *(a) Fraction of all hydrogen in the H-rich phase for a 3 wt% parcel cooling at 2, 5 and 10 GPa, in the equilibrium limit (solid) and the delayed limit (dashed). (b) Fraction withheld at the spinodal as a function of local hydrogen mass fraction. The curves for the three pressures coincide.*

follows from the separable form of Eq. (2.4) (Appendix A); it is not independent evidence.

3.3 Audit of the Published Fit against its Source Data

Our implementation reproduces the published model (Section 2.6), so any mismatch with the source simulations belongs to the published fit itself. The fit agrees with all ten one-phase runs. For the two-phase run at 1.65 GPa, its gap at 3,500 K lies entirely above the bulk composition of the cell, a miss of 35 K in temperature. It reproduces both compositions within the quoted ± 0.05 for only one of the four two-phase runs (Table B.1, Figure 3.3b).

Near-critical runs. The largest misses occur for the two runs closest to the critical curve. The published T_C lies 26 K above the run at 1.65 GPa and 3,500 K, and 157 K above the run at 5.14 GPa and 3,000 K. For the first run the model predicts a gap centred at $x \approx 0.71$, whereas the simulation shows a gap centred at 0.52. For exactly these two runs the source table and the source figure disagree. The digitized Figure 2a places the pairs at (0.419, 0.629) and (0.514, 0.736), whereas ED Table 1 lists (0.458, 0.576) and (0.587, 0.706). The differences range from -0.073 to $+0.053$. For the two wide-gap runs, table and figure agree to within 0.0006.

In all four two-phase runs, the per-phase atom numbers in ED Table 1 add up to the full simulation cell, so every interfacial atom is counted in one of the two phases. With sharp interfaces this matters little. Near T_C, however, interfaces broaden, and such a count pulls both compositions toward the bulk composition of the cell, which is 0.526 and 0.629 for these runs. The two composition estimators described in the source methods [13] could therefore differ most for exactly these runs, as observed. We treat the near-critical compositions as uncertain at the 0.04–0.07 level. This is an inference from the published numbers, not a demonstrated cause.

Homogeneous simulations. The homogeneous simulations reveal a second mismatch (Figure 3.3c–e). The published fit predicts an excess volume of $-1.94\,\mathrm{cm^3\,mol^{-1}}$ at $x = 0.588$, whereas the simulations give $-0.94 \pm 0.16\,\mathrm{cm^3\,mol^{-1}}$. Because the excess volume sets the pressure dependence of T_C, the value $\pi = -35$ GPa reproduces the critical curve between 2 and 10 GPa, but the directly simulated volumes imply a pressure scale about twice as large in magnitude. The fit also underestimates the excess enthalpy at $x = 0.24$. In total, the published fit has $\chi^2 = 0.4$ for the 14 phase states, 71.3 for the eight compositions and 140.0 for the nine homogeneous properties.

These comparisons reproduce checks against the data from which the fit was built. They are not independent validation, and they do not show that any simulation is wrong. They show that one four-parameter model cannot follow all of the source data at once.

3.4 Recalibrated Metastable Interval

Table 3.2 and Figure 3.4 summarize the nine refits. Refitting the published form (class M1) to all data lowers χ^2 from 212 to 60. The best fit is much less asymmetric than the published one (Table C.1) and has a lower critical temperature, $T_C(2\,\mathrm{GPa}) = 3223$ K (68% range 3201–3248 K). It pays for this with the phase state of the run at 1.65 GPa and 3,500 K, which it places 264 K above its binodal. The metastable interval of the baseline parcel shrinks to 92 K (69–120 K), and the withheld fraction to 49%.

The data treatments show which data drive this shift (Table C.1). Without the homogeneous data, the fit returns to the published asymmetry, $B/A = 6.4$ against 5.65, but with interaction parameters three

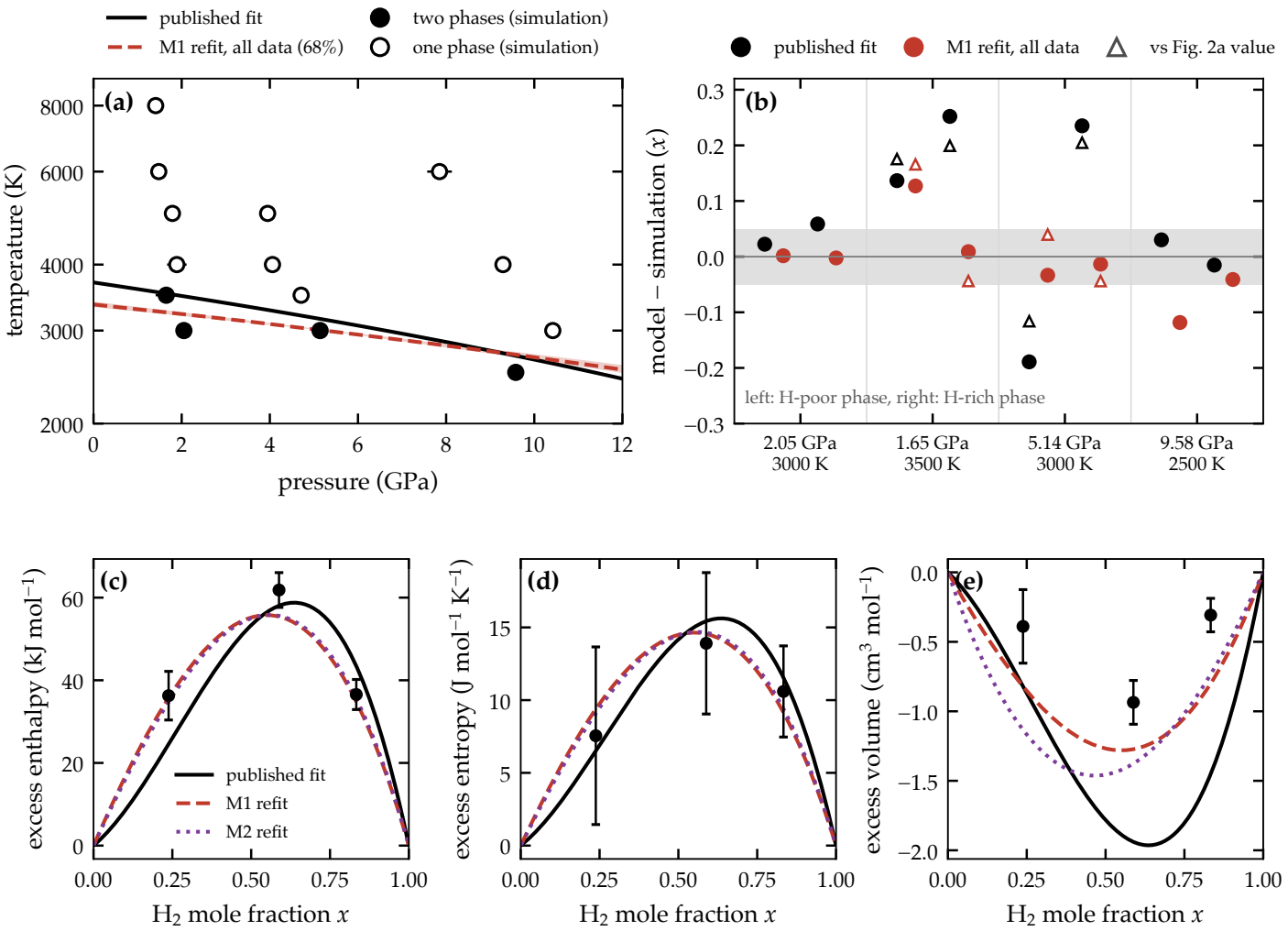


Figure 3.3: ***Audit of the published fit against its source simulations.*** *(a) Phase state of the 14 coexistence runs (filled: two phases; open: one phase) with the published critical curve (black) and the critical curve of the refit to all data (red, 68% band). (b) Model minus simulation for the eight coexisting compositions, against ED Table 1 (circles) and the digitized Figure 2a (triangles); the grey band is the quoted ±0.05. (c–e) Excess enthalpy, entropy and volume of the homogeneous simulations at 5 GPa and 4,000 K (symbols, 1σ), with the published fit and the refits of classes M1 and M2.*

times larger and a lower τ. Its interval is 134 K, not 414 K, so the asymmetry alone does not set the interval: the excess entropy and the strength of the interaction matter as much. With the homogeneous data but without the compositions, the excess is nearly symmetric, $B/A = 1.4$, and the interval is 67 K. The compositions themselves discriminate weakly between these fits: their misfit is $\chi^2_{\mathrm{comp}} = 11.7$ for the strongly and 13.3 for the weakly asymmetric fit. The difference lies mainly in the phase states and the homogeneous data. Using the Figure 2a compositions instead of ED Table 1 gives 95 K, and dropping the excess volumes gives 86 K: the volume mismatch controls the pressure dependence of T_C, but hardly the interval at 2 GPa. Giving the two near-critical runs a composition uncertainty of 0.15 raises the asymmetry to $B/A = 2.5$ and lengthens the interval to 242 K.

Every refit that uses the coexisting compositions places the run at 1.65 GPa and 3,500 K between 95 and 270 K above its binodal. This reflects a tension within the source data. The narrow gap at 5.14 GPa and 3,000 K requires T_C close to 3,000 K at that pressure, whereas the two-phase state at 1.65 GPa requires $T_C > 3{,}500$ K. A linear critical curve meets both only if it falls by more than about 143 K per GPa. The published fit falls by 106 K per GPa, and the fit to phase states and homogeneous data by 52 K per GPa.

The more flexible classes fit the data with $\chi^2 = 50$ (M2), 53 (M3) and 52 (M4), against 60 for M1. Their intervals are 41, 89 and 161 K. None of the classes fits all data within the quoted errors: for 31 data points, χ^2 stays between 50 and 60. Across all nine refits the median interval ranges from 41 to 242 K, compared with 414 K for the published fit (Table 3.2, Figure 3.4a). For the most flexible class, M4, the 95% range is 62–479 K, and 4.9% of its samples reach the published value.

The largest split-chain $\hat{R}$ over all parameters and de-

Table 3.2: ***Recalibration of the hydrogen–silicate mixing model.*** *χ^2 terms at the best fit for the 14 phase states, the 8 compositions and the 9 homogeneous properties, and posterior medians with 68% ranges of the critical temperature at 2 GPa, the metastable interval and the withheld hydrogen fraction for a 3 wt% parcel at 2 GPa. Fits marked † leave out data (dashes), so their χ^2 cannot be compared with the others.*

Fit	χ^2_{phase}	χ^2_{comp}	χ^2_{hom}	T_C (K)	ΔT_{ms} (K)	f_{H} (%)
Published fit	0.4	71.3	140.0	3489	414	65
M1, all data	22.7	13.3	23.5	3223 [3201–3248]	92 [69–120]	49 [45–52]
M1, Fig. 2a compositions	20.2	19.3	23.5	3243 [3218–3272]	95 [71–124]	49 [45–52]
M1, $\sigma_x = 0.15$ near T_C	2.9	20.0	19.4	3422 [3359–3485]	242 [176–317]	59 [55–62]
M1, without volumes†	10.4	13.7	2.5	3298 [3276–3318]	86 [64–112]	49 [46–52]
M1, phases + compositions†	9.0	11.7	–	3311 [3288–3328]	134 [86–202]	65 [61–68]
M1, phases + homogeneous†	0.4	–	11.5	3471 [3417–3533]	67 [26–124]	45 [36–51]
M2, all data	23.8	5.3	20.6	3209 [3184–3237]	41 [22–65]	40 [35–45]
M3, all data	22.0	15.3	15.6	3236 [3207–3270]	89 [67–115]	41 [38–44]
M4, all data	21.1	17.3	14.1	3266 [3227–3327]	161 [98–284]	48 [42–55]

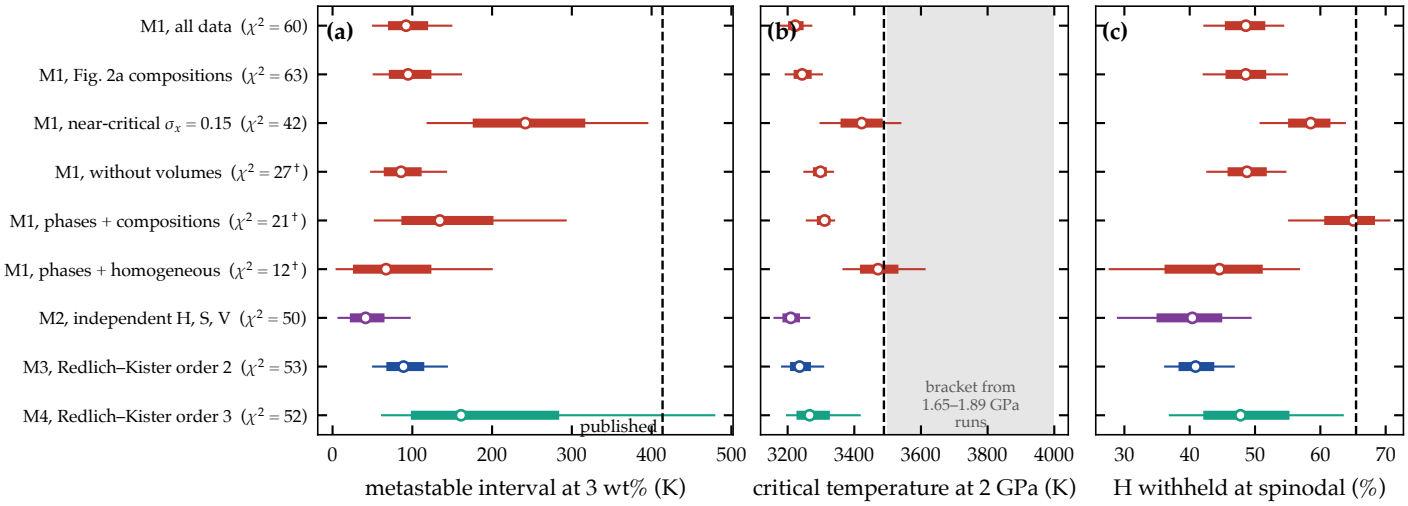


Figure 3.4: ***Posterior distributions of the metastable interval, critical temperature and withheld hydrogen.*** *(a) Metastable interval, (b) critical temperature at 2 GPa and (c) withheld hydrogen fraction for a 3 wt% parcel at 2 GPa, for each model class and data treatment. Thick bars show 68% and thin bars 95% intervals; open circles mark medians. Dashed lines mark the published fit; the grey band in (b) is the bracket set by the two-phase run at 1.65 GPa and the one-phase run at 1.89 GPa, drawn without correction to 2 GPa. Labels give the best-fit χ^2; fits marked † use fewer data.*

rived quantities is 1.016, for B in the M1 fit to phase states and compositions; $1/\tau$ in that fit reaches 1.014. Of the nine fits, 1 exceeds the threshold of 1.01 recommended for the rank-normalized statistic [36], so the posterior quantiles of this fit are slightly less reliable than those of the others. In every other fit $\hat{R}$ stays at or below 1.005.

3.5 Identifiability of the Spinodal

The perturbation of Eq. (2.13) leaves the binodal, the equilibrium free energy and every homogeneous state outside the gap unchanged; we verified this to within 4×10^{-12} J mol^{-1}. It nevertheless moves the spinodal (Figure 3.6). At the largest admissible amplitude, $\lambda = 0.22$, the 3 wt% spinodal rises by 16.2 K, the 5 wt% spinodal falls by 4.9 K and the 2 wt% spinodal rises by 124.4 K. Coexistence data alone therefore cannot identify the spinodal. Any spinodal inferred from them is a property of the assumed functional form.

3.6 Heat of Hydrogen Exsolution

Mixing hydrogen into silicate melt absorbs heat: the excess enthalpy of the source simulations is positive at every composition [14]. Separation therefore releases heat. We obtain this heat from the same free energy that sets the binodal, by automatic differentiation in Atmodeller (Appendix A). The relevant quantity for a planet is the partial molar enthalpy of H_2 in the melt, $\bar{h}_{H_2}$: the heat released when a small amount of hydrogen leaves the melt and ends as pure H_2. For the published fit, $\bar{h}_{H_2}$ is 51–64 MJ per kg of hydrogen between 1 and 3 wt% and 2 and 5 GPa. The class-M1 refit posteriors that include the en-

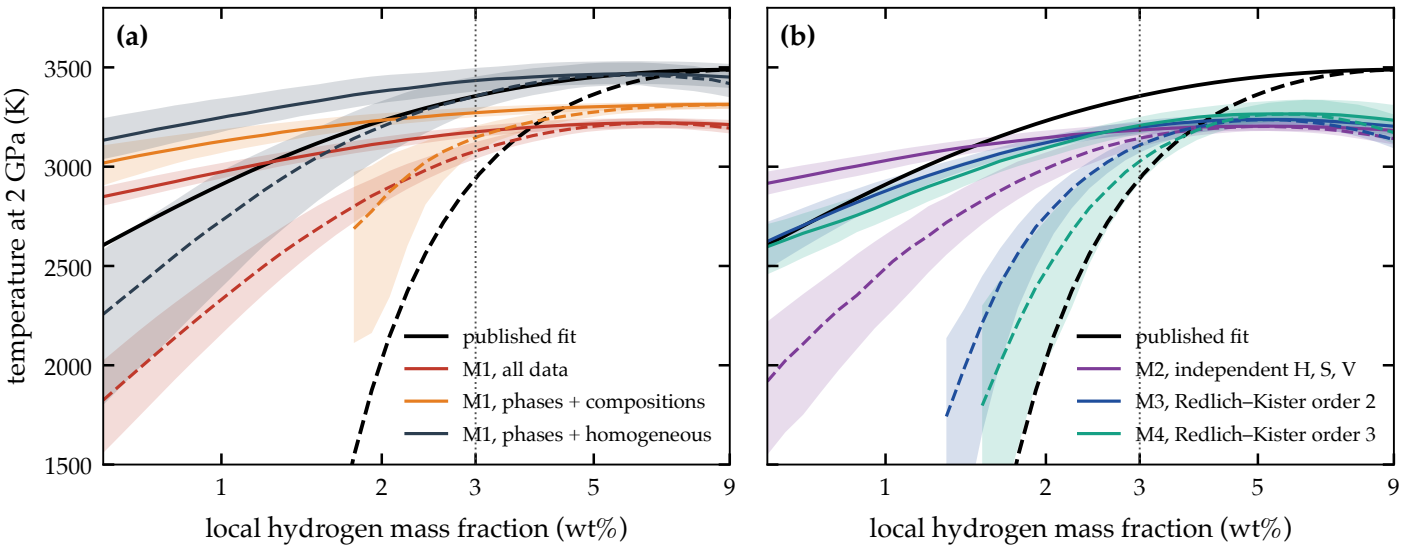


Figure 3.5: ***Posterior binodals and spinodals at 2 GPa.*** *Medians (lines) and 68% bands of the binodal (solid) and spinodal (dashed) for (a) class M1 with three data treatments and (b) classes M2–M4 with all data, compared with the published fit (black).*

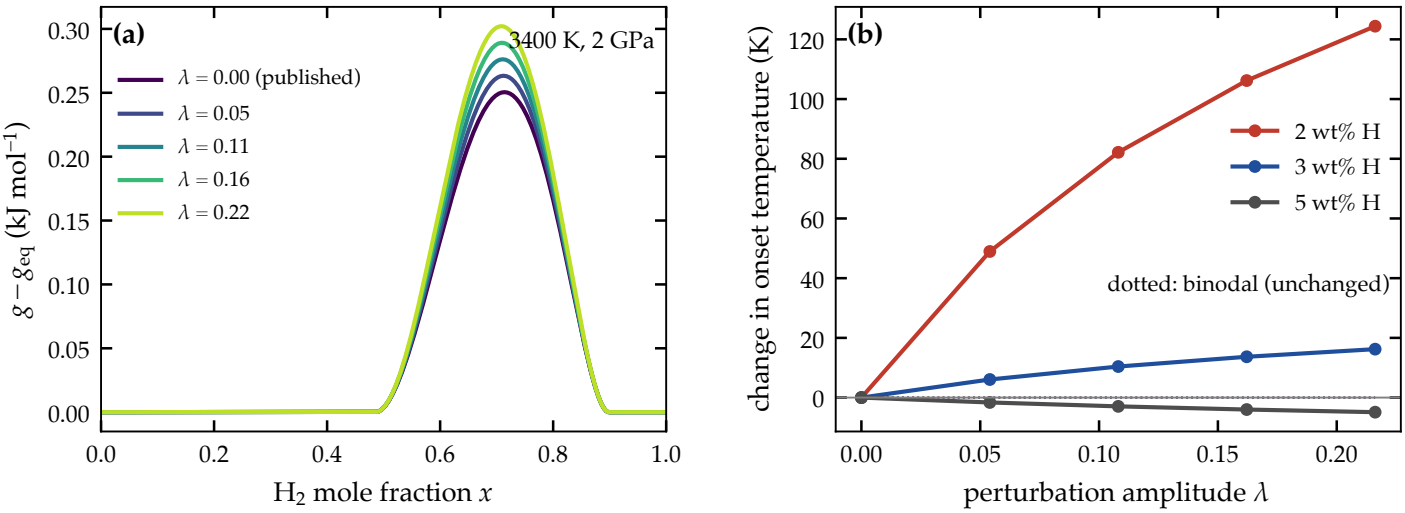


Figure 3.6: ***Two free energies with the same equilibrium but different spinodals.*** *(a) Homogeneous Gibbs energy above the equilibrium (convex-envelope) free energy at 3,400 K and 2 GPa for perturbation amplitudes $\lambda = 0$ to 0.22, the largest amplitude that keeps every heat capacity positive. Outside the gap all curves are zero. (b) Change of the spinodal (solid) and binodal (dotted) temperatures with λ for 2, 3 and 5 wt% local hydrogen.*

thalpy data give 35–93 MJ kg^{-1} (95% intervals over the same grid), with medians of 72 and 43 MJ kg^{-1} at 1 and 3 wt% and 2 GPa. In refit M1 ($B/A = 1.5$, so $B < 2A$) the heat is largest for dilute hydrogen, where each H_2 molecule is surrounded by silicate. The published fit ($B/A = 5.65$) instead peaks at $x = 0.26$ ($w = 1.40$ wt%), where W'' changes sign, and falls to 35 MJ kg^{-1} at infinite dilution and 2 GPa (Appendix A.7). The 1 wt% values ($x = 0.20$) lie below the most hydrogen-poor simulation ($x = 0.24$, 1.2 wt%), so they are extrapolations.

The source data favour the refits here. At the most hydrogen-poor simulated mixture ($x = 0.24$, 5.08 GPa, 4000 K), the simulated excess enthalpy is 36.3 ± 5.9 kJ mol^{-1} (1σ, after the endmember pressure correction; Table B.2). The published fit gives 24.5 kJ mol^{-1}, 2.0 standard deviations too low, and refit M1 gives 35.4 kJ mol^{-1}. At $x = 0.83$ the published fit is too high instead (43.5 against 36.6 ± 3.6 kJ mol^{-1}; M1: 35.1), so its fault is the asymmetry of the excess. The refit to phase compositions alone, without enthalpy data, predicts 75 kJ mol^{-1} at $x = 0.24$ and 100–298 MJ kg^{-1}; the enthalpy data rule it out.

Per kilogram of hydrogen, this heat is 48–127 times the latent heat of melting per kilogram of silicate (0.73 MJ kg^{-1}; [35]). Hydrogen makes up only 1–3% of the mass, however, so per kilogram of interior the two heats are comparable. Over the first 300 K below the binodal, a 3 wt% interior at 2 GPa (refit M1, Section 3.9) releases 1.4 (1.1–1.7) MJ per kg of interior, 1.9 times the latent heat. At 1 wt% it releases 0.5 (0.4–

0.7) MJ kg^{-1}, 0.7 times the latent heat.
The data constrain the heat only at 4000 K. Consider an excess heat capacity $c_{ex} = \pm 50\, x(1-x)$ J mol^{-1} K^{-1} per oxide-basis mole, at most ± 12.5 at $x = 0.5$, with the excess enthalpy held at its simulated value at 4000 K. At 3000 K it changes $\bar{h}_{H_2}$ by $\mp 50\,(1-x)^2$ kJ mol^{-1}, lowering it for positive c_{ex}. That is 14–27% of the published and M1 median values over the same grid, and the shift is largest for dilute hydrogen.

3.7 How Far Below the Binodal Separation Starts

The delayed-exsolution window of Section 3.1 assumes that a cooling parcel stays homogeneous until the spinodal. A metastable parcel can instead nucleate the hydrogen-rich phase. We compute the critical nucleus in square-gradient theory [6, 7], using the same free energy, a gradient coefficient κ and a constant molar volume. The barrier scales exactly as $\kappa^{3/2}$, tends to classical theory near the binodal and vanishes as the 3/2 power at the spinodal (Appendix A). Nucleation becomes effective once the barrier falls to $W^*/k_BT = \ln(J_0 V t)$ [17]. We take 69–123. The lower value uses $J_0 = 10^{30}$ m^{-3} s^{-1}, $V = 1$ m^3 and $t = 1$ s. The upper value uses $J_0 = 10^{40}$ m^{-3} s^{-1}, $V = 1$ m^3 and $t = 1$ Myr. A volume of 10^{20} m^3 over 1 Gyr would raise the criterion to 153–176, which shortens the delays further. The molar volume is 18.2 cm^3 per mole of mixing units, varied by $\pm 30\%$. It is the cell volume of the two-phase run at 2.05 GPa and 3000 K in ED Table 1: 258 atoms of 13.52 Å^3 over 115.5 mixing units.
The gradient coefficient follows from a simulated two-phase profile. Figure 1c of [13] shows the time-averaged number densities of H, O and Si across the interface at 10 GPa and 2500 K, below the published T_C of 2643 K there. We fitted their equation 4 to each species and took the 10–90% width, $2\,\mathrm{artanh}(0.8)\,\delta$, which is 0.28–0.33 nm. If all species share δ and the centre, the composition x has the same width, so we use this range for x. Each model's free energy at that state converts the width to κ (Appendix A.8). We then hold κ fixed at the parcel's pressure and temperature. Capillary waves and drift of the slab can only broaden a time-averaged profile, so any broadening makes κ and the delays below too large.
With this width, a 3 wt% parcel at 2 GPa nucleates 1.2–2.9 K below the binodal for the published fit, and 0.2–1.6 K below it across the M1 posterior (molar volume $\pm 30\%$). At most 1.3% of the hydrogen is withheld. A wider profile from a larger cell of [39] (0.5 nm at 2.5 GPa and 3000 K; [24], Fig. 5.1) gives 7.3–14.1 K

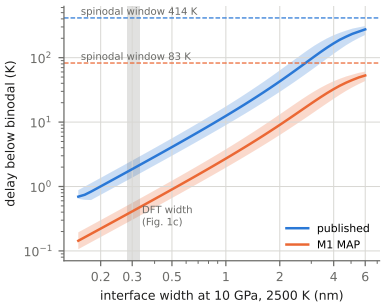


Figure 3.7: ***Nucleation ends the metastable delay close to the binodal.*** *Undercooling below the binodal at which a 3 wt% parcel at 2 GPa nucleates, against the 10–90% interface width at 10 GPa and 2500 K, for the published fit (blue) and refit M1 at its MAP (orange). Solid lines are band medians. Bands span $W^*/k_BT = 69$–123 and molar volume $\pm 30\%$. Dashed lines mark each binodal–spinodal window (414 K published, 83 K M1). The grey band is the width of the simulated profile.*

and at most 5.8% withheld. Reaching half of the published window would need an interface 3.8–5.6 nm wide at 10 GPa and 2500 K, 11–20 times the simulated width. Refit M1 needs about the same width, 3.9–5.8 nm, for half of its own window (Figure 3.7).

3.8 Cooling Paths of a Closed Parcel

Figure 3.8 follows a closed 3 wt% parcel at 2 GPa as it loses heat. Sensible heat comes from the Atmodeller end members, liquid $MgSiO_3$ (1.46 kJ kg^{-1} K^{-1}) and ideal-gas H_2 (18.4 kJ kg^{-1} K^{-1} at 3000 K). They are weighted by mass, without a pressure correction. Hydrogen then supplies 28% of the sensible heat capacity of a 3 wt% parcel. For comparison, [30] take 1.195 kJ kg^{-1} K^{-1} for the silicate melt. If the parcel stayed homogeneous to the spinodal and then separated, it would release 1.2 MJ kg^{-1} at once and reheat by 286 K (refit M1: 0.50 MJ kg^{-1}, 64 K). Limited by nucleation, the parcel separates 2.1 K below the binodal, releases 12 kJ kg^{-1} and reheats by only 1.6 K. Its path is then indistinguishable from equilibrium. All paths reach the same final state after the same heat loss (3.42 MJ kg^{-1} to 2700 K): delay changes when heat is released, not how much.

3.9 A Convecting Interior Releases Exsolution Heat

In a convecting interior every parcel reaches the top, where the melt meets the envelope. The envelope already provides the hydrogen-rich phase, so no nucleation is needed there. Deeper layers are hotter and, because the critical temperature falls with pressure, more miscible. The interior composition therefore follows the hydrogen-poor binodal at the top, and

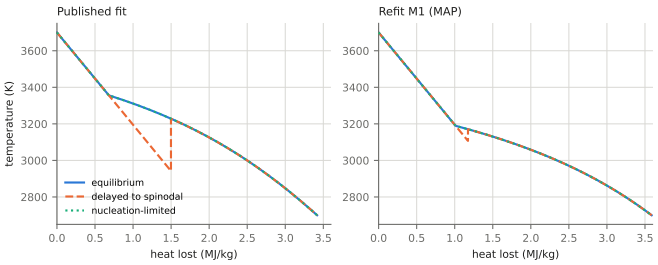


Figure 3.8: ***Cooling paths of a closed 3 wt% parcel at 2 GPa.*** *Temperature against heat lost from 3700 K for the published fit (left) and refit M1 at its MAP (right). Solid lines show equilibrium separation. Dotted lines show nucleation-limited separation, which overlies the equilibrium path. Dashed lines show a mean-field bound that nucleation pre-empts: separation delayed to the spinodal, which would reheat the parcel by 286 K (published fit) or 64 K (refit M1).*

the exsolved hydrogen joins the envelope.

Over the first 300 K of cooling below the binodal, a 3 wt% interior with refit M1 at 2 GPa loses 79% (72–85%) of its hydrogen and releases 59 MJ per kg of it. This heat is 2.3 (1.9–2.9) times the sensible heat lost over the same interval by the interior and the hydrogen it has exsolved, which we let cool with the top of the interior; for 1 wt% it is 1.1 (0.9–1.4) times (published fit, 3 wt%: 1.8). Without the exsolved hydrogen in the sensible heat, the ratio at 3 wt% is 2.8 (2.2–3.5). The ratio also depends on the chosen interval. About half of the exsolution heat of the first 300 K is released in the first 100 K, while the sensible heat grows in proportion to the cooling. The median ratio is therefore 3.8 over the first 100 K and 1.5 over the first 600 K; for the published fit it is 2.6 and 1.2.

Just below the binodal, exsolution adds up to 6.0 (4.2–8.9) times the sensible heat capacity at 3 wt% (Figure 3.9b). The effective heat capacity of the interior there is thus up to 7.0 (5.2–9.9) times the sensible one, and the interior cools markedly more slowly while it exsolves.

These numbers rest on five assumptions, and each one shifts them. First, overturn is fast compared with cooling, so the interior stays uniform. Second, the top pressure stays at 2 GPa. In the evolution of [30], the binodal pressure rises from 0.9 to 4.9 GPa while its temperature falls by 925 K. At higher pressure the melt at the top dissolves more hydrogen. A rise at that rate lowers the exsolution heat over the first 300 K by 15% for refit M1 (median) and by 28% for the published fit. Third, the whole interior cools by as much as its top. On an adiabat the mean temperature falls faster than the top temperature, and the ratio to sensible heat falls by the same factor. A factor of 1.3, for example, lowers the median ratio from 2.3 to 1.8. Fourth, the model includes no gravitational energy from rising hydrogen or settling melt; in [30] that energy is part of the gravitational term. Fifth, the exsolved hydrogen ends as pure H_2.

Near the binodal the coexisting hydrogen-rich phase still carries much silicate. In refit M1 it holds only 14 wt% hydrogen 1 K below the binodal (median). Suppose the hydrogen leaves the melt as that phase. Each kilogram of hydrogen then releases only 48% of the pure-H_2 heat. The departing phase also removes silicate, so more hydrogen must leave per kelvin of cooling. The heat released per kelvin is then 59% of the pure-H_2 value 1 K below the binodal and 75% at 100 K below it. Exsolution then adds at most 3.5 (2.7–4.7) times the sensible heat capacity, and the ratio over 300 K falls to 1.7 (1.4–2.1). If the hydrogen instead joins an envelope that is already at the coexisting composition, it releases 84% of the pure-H_2 heat 1 K below the binodal, and exsolution adds at most 5.0 (3.7–6.9) times the sensible heat capacity. The remainder appears higher in the envelope as the silicate rains out (Section 4.1). The interior cools more slowly in every case, but the pure-H_2 factors are upper bounds for the interior itself.

At the scale of the whole planet the term is smaller. We digitized the fiducial evolution of [30] (6 $M_\oplus$ with 3% hydrogen; their Figures 5 and 6). Along it we integrate our partial molar internal energies, the enthalpies minus the pressure–volume term, which is the energy basis of their budget. The planet exsolves 57% of its hydrogen between 5 Myr and 9 Gyr. Across the class-M1 refits that use the enthalpy data, the heat is 4–7% of the energy the planet radiates over that time and 5–7% of its gravitational energy release, which supplies most of the luminosity. It is 0.8 (0.6–1.0) times the thermal-energy loss in the same budget, and 0.8 (0.7–1.0) times the sensible heat the interior loses as it cools with the binodal. The estimate is a scaling, not a prediction. The thermal history comes from the 2025 phase diagram. Its binodal path falls below the critical temperature of the 2026 fit only after about 5 Gyr, when 96% of that exsolution has already happened. For refit M1 fitted to all data it stays above the critical temperature throughout, but 40% of the samples of all M1 fits with enthalpy data fall below it. A consistent planetary estimate needs an evolution model that uses the same free energy for both. No evolution model we checked includes the heat of exsolution [29, 30, 34, 35, 38].

The Atmodeller solver reproduces the mixing enthalpy along one published-fit parcel path to 9×10^{-16} over 80 solves (Appendix D). The interior paths and the nucleation onsets are not replayed through the solver.

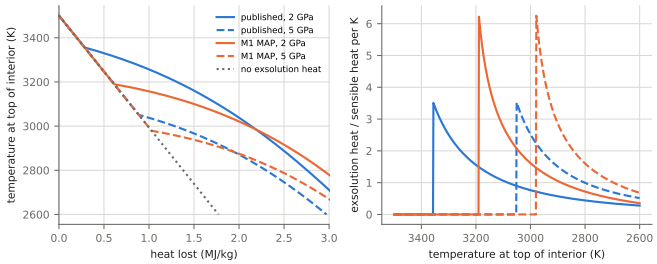


Figure 3.9: ***Exsolution heat in a convecting 3 wt% interior.*** *Published fit (blue) and refit M1 at its MAP (orange), with the top at 2 GPa (solid) and 5 GPa (dashed). (a) Temperature at the top of the interior against heat lost per kg of system. The dotted line is the sensible heat alone, which is the same for all four cases. (b) Exsolution heat per kelvin of cooling divided by the sensible heat capacity. The effective heat capacity is one plus this ratio.*

4. Discussion

The Results give three findings of different weight. Hydrogen exsolution releases a large heat per kilogram, and the simulated excess enthalpies constrain it. Homogeneous nucleation keeps exsolution close to equilibrium, so the metastable window of the published fit is not reached. The source data constrain the heat, but they do not fix the spinodal. This chapter discusses each finding and then the data that would sharpen them.

4.1 What Sets the Heat of Exsolution

The heat follows from the excess enthalpy, which the homogeneous source simulations measure directly [14]. In refit M1, which matches the simulated enthalpies, it is largest for dilute hydrogen, where each H_2 molecule is surrounded by silicate. The published fit instead peaks near 1.40 wt% (Section 3.6). Below 1.2 wt%, the most hydrogen-poor simulation, both are extrapolations. The binodal and the spinodal also depend on the excess entropy and on the free energy inside the gap. The heat is therefore the better constrained quantity: it depends on the enthalpy of homogeneous states, which the simulations sample.

The published fit underestimates the simulated excess enthalpy of the most hydrogen-poor mixture, 24.5 against 36.3 ± 5.9 kJ mol^{-1} (1σ), which is 2.0 standard deviations low. It overestimates the excess at $x = 0.83$. The refit M1 to all data gives 35.4 kJ mol^{-1} at $x = 0.24$. A refit to the phase compositions alone predicts 75 kJ mol^{-1}, which the enthalpy data exclude. A fit to phase data can thus reproduce the gap with the wrong enthalpy. An evolution model that takes energies from a mixing model, and not only phase boundaries, needs a model fitted to the enthalpies as well.

Two limits apply. First, the enthalpy data exist only at 4000 K. The melt is reactive, and the source simulations form hydroxyl, water and reduced silicon species [13]. Their equilibria give an excess heat capacity, which all our model classes set to zero. Section 3.6 estimates how much a plausible excess heat capacity changes the heat at 3000 K. Second, near the critical point the hydrogen-rich phase is not pure H_2 and carries silicate. Less heat is then released where hydrogen leaves the melt, and the rest follows higher in the envelope as the silicate rains out [35]. Our interior model sends the exsolved hydrogen to pure H_2. It gives the total heat along the path, not where in the planet that heat appears. Section 3.9 shows how much less heat the other end states release at the top of the interior.

4.2 Exsolution Heat and Planetary Cooling

In a convecting interior the heat matters most while the top of the interior exsolves. Over the first 300 K below the binodal, the released heat is 2.3 times the sensible heat at 3 wt% (refit M1). Just below the binodal, exsolution adds up to 6.0 times the sensible heat capacity (Section 3.9). At a given heat flow, the top of the interior therefore takes 3.3 times as long to cool through these 300 K, and it cools up to 7.0 times more slowly just below the binodal. At 1 wt% the ratio is 1.1, so the effect grows with the hydrogen content of the melt. These factors assume that the hydrogen leaves as pure H_2, so they are upper bounds (Section 3.9).

For the planet as a whole, the term is smaller. In the energy budget of Rogers et al. [30], gravitational contraction supplies most of the radiated energy. The sensible heat of the interior is a minor term. Against that budget, the exsolution heat is 4–7% of the energy radiated over 9 Gyr and about 0.8 times the thermal-energy loss (Section 3.9). If the planet kept the same luminosity at each stage, the extra heat would lengthen its evolution by 4–6% (published fit: 4%; best-fitting refit: 5%). This estimate mixes two phase diagrams. The thermal history of Rogers et al. uses the 2025 preprint fit [33]. Its binodal path stays above the 2026 critical temperature until about 5 Gyr, while our enthalpies come from the 2026 model. A consistent planetary value needs an evolution model that takes the phase boundary and the heat from the same free energy. None of the five evolution models we checked includes the heat of exsolution, either explicitly or through a mixture equation of state [29, 30, 34, 35, 38]. In the last of these, the mantle holds no hydrogen.

4.3 Exsolution Stays Close to Equilibrium

Both routes to separation keep it close to the binodal. At the top of a convecting interior, the melt touches the envelope, which is already the hydrogen-rich phase. Hydrogen can leave there without nucleation. Deeper fluid is hotter and, because T_C falls with pressure, further from the gap. It enters the gap only when convection carries it to the top. Away from the envelope, a parcel must nucleate the new phase. With the interface width of the simulated profile, homogeneous nucleation ends metastability 1.2–2.9 K below the binodal for the published fit and 0.2–1.6 K for refit M1. At most 1.3% of the hydrogen is withheld (Section 3.7). The wider profile of a larger cell gives 7.3–14.1 K. Nucleation on interfaces, such as crystals or droplets of an earlier-separated phase, can only shorten this delay [31].

The binodal–spinodal window of the published fit, 414 K for this parcel, is therefore a mean-field upper bound that a cooling parcel does not reach. Young et al. [39] plotted this window for the 2025 fit (their Fig. 2). Our calculation shows how far nucleation falls short of it. Reaching half of the window would need an interface 11–20 times wider than the simulated one (Section 3.7). The spinodal itself is a mean-field construct. For short-range forces, a sharp spinodal exists only in mean-field theory [19], and the crossover from nucleation to spinodal decomposition is gradual [2].

Three approximations limit these delays. The gradient coefficient comes from one simulated profile in a small cell, which may include capillary broadening. Broadening makes the coefficient and the delays too large, so removing it would shorten the delays further. How the coefficient transfers between pressures matters for refit M1 but hardly for the published fit. Matching the simulated width at 2 GPa instead of 10 GPa shortens the M1 delays about threefold (two- to fivefold across the posterior) and the published-fit delays by 5–6%. The convention used here gives the longest delays. The model uses one composition variable and one molar volume for both phases. Larger volumes and longer times raise the nucleation criterion and shorten the delays further.

With near-equilibrium exsolution, the case for delayed hydrogen release largely disappears. A lag would leave the envelope with less hydrogen than equilibrium predicts, not more. It would therefore raise the mean molecular weight of the atmosphere for a while. In a convecting interior the top degasses without delay, and an isolated parcel lags by at most 2.9 K for the simulated interface width. Effects of a lag on the atmosphere are therefore negligible. A longer delay would need two conditions together. The interior would have to be stratified in composition, so that fluid enters the gap without touching the hydrogen-rich phase. And the nucleation barrier would have to be much larger than the simulated interface implies.

4.4 What the Source Data Can and Cannot Constrain

The binodal and the spinodal probe different parts of the free energy. The binodal follows from the common tangent, so it depends only on the free energy at the two coexisting compositions. The spinodal depends on the curvature of the homogeneous free energy between them, a state that phase-separated simulations do not sample. The perturbation of Section 3.5 shows this for the present system. It changes the curvature inside the gap and leaves every bulk equilibrium quantity unchanged. That bulk equilibrium fixes only the convex envelope of the free energy is textbook thermodynamics. The perturbation is one admissible counterexample, so its shifts are lower bounds on what the data leave open. With every heat capacity kept positive, it moves the 3 wt% spinodal by only 16.2 K. That is small against the M1 68% range of the baseline window, 69–120 K. It moves the 2 wt% spinodal, near the cold end of the gap, by 124.4 K. At the baseline, the larger uncertainty comes from the choice of functional form. The same limit applies to spinodals inferred from solubility data in other silicate–volatile systems [27].

Fitting a smooth model with few parameters ties the curvature inside the gap to the free energy outside it. The spinodal is then as well constrained as these parameters, and three of them matter together: the asymmetry B/A, the excess entropy set by τ, and the strength of the interaction. The source data pull them in different directions. Without the homogeneous data, the fit returns to the published asymmetry ($B/A = 6.4$). The homogeneous enthalpies favour a weak asymmetry ($B/A = 1.4$). None of our refits reproduces the published combination. Their median windows span 41–242 K, against 414 K for the published fit. In mean-field theory, near the critical point, the spinodal lies three times as far below T_C as the binodal at the same distance from x_C [9]. Near-critical data therefore constrain the spinodal of near-critical parcels, but not of a parcel at $x = 0.435$. The runs most informative about the curvature are also those whose compositions are least certain (Section 3.3).

Two further effects bias the source data in ways that the posteriors do not include. First, the coexistence

runs use cells of 255–349 atoms with edges of 11–12 Å [14]. No size test exists for H_2–$MgSiO_3$, and Young et al. ran a single cell size [39]. The closest analogue is NH_3–H_2. There, cells of this size pull the coexisting compositions toward the bulk by 0.1–0.2, and they converge only above about 8,000 atoms ([15], Fig. S2). Correcting an inward bias of 0.05–0.10 per phase would raise the best-fit T_C at 2 GPa by about 5–35 K. It would change the 3 wt% window of the best fit by -50 to $+45$ K. A bias on the hydrogen-poor side narrows the window, and one on the hydrogen-rich side widens it. A symmetric bias changes it by at most 15 K. Cell size may also change whether a run near T_C shows one phase or two, in either direction. Wang, Hamel & Cheng found H/He demixing temperatures about 2000 K below earlier ab initio studies with 64 atoms or fewer [37]. They name cell size as the likely main cause, by ruling out other error sources, but they did not compute small-cell demixing temperatures. Their result does not transfer to direct-coexistence runs. In their own structure-factor method, small cells push toward mixing, which is the opposite sign.

Second, the runs are not strictly binary. In the coexistence runs the hydrogen-rich phase is enriched in silicon, and near T_C the tie lines lie well off the H_2–$MgSiO_3$ join [14]. Stability then requires the full composition Hessian to be positive definite. Its smallest eigenvalue can vanish before the curvature along the H_2–$MgSiO_3$ join does. For a given free energy, the binary spinodal is then a lower bound in temperature. The binodal can move as well, so the sign of the change in the window is not fixed.

The excess volume adds a smaller caveat. The same group has simulated homogeneous $MgSiO_3H_4$ with the same functional [25]. These data put the excess volume at 5 GPa and 4000 K between -1.4 and $+0.2\,\mathrm{cm^3\,mol^{-1}}$, smaller in magnitude than the published $-1.78\,\mathrm{cm^3\,mol^{-1}}$. Their smooth equation-of-state fit gives the published value, but it misses their two 4000 K points nearest 5 GPa by about 11 and 26 standard deviations in pressure. They also find an excess volume that varies with pressure, which a constant π cannot follow. A π fitted to the critical curve between 2 and 10 GPa can then differ from the volume at 5 GPa without any inconsistency in the data. The mismatch of Section 3.3 therefore holds at one pressure and temperature, against the PBEsol endmembers of the source data.

4.5 Data That Would Sharpen Both Results

The cheapest improvement uses data that already exist. Per-run composition estimators, time series and interface widths from the source authors would sharpen both results. The interface widths fix the gradient coefficient, and with it the nucleation delay. The estimators and time series fix the compositions of the near-critical runs, which drive T_C and the spinodal. The interfacial tension, from the stress anisotropy of the existing coexistence runs, would give an independent value of the gradient coefficient. Homogeneous enthalpies below 4000 K would test the temperature dependence of the heat.

New simulations would narrow the spinodal only a little. A screen over candidate measurements uses the posterior samples of the four model classes, each fitted to all data, with equal weight (Figure 4.1). Consider one new homogeneous simulation at $x \approx 0.35$, with the precision of the existing ED Table 2 entries. It would reduce the posterior standard deviation of the baseline spinodal only from 80 K to 72 K. The pooled spread has two main sources. The four model classes disagree on the mean spinodal, and class M4 alone has a wide posterior. One more homogeneous point narrows each class by only a few kelvin, so it removes neither source.

A direct measurement of the curvature does only slightly better. At 2 GPa and 3600 K, a 10% measurement of $\partial^2 g/\partial x^2$ at the baseline composition would reduce the standard deviation from 80 K to 64 K. The screen covers 3200, 3400 and 3600 K. Of these, 3600 K is the only temperature at which every posterior sample is one phase (100%); at 3400 K a single M4 sample is already two-phase. In a one-phase parcel the simulation cannot decompose during the run. For the same reason, the measured state lies outside the miscibility gap, where the perturbation of Section 3.5 and its first two derivatives vanish. 3600 K is also above the critical temperature of every sample and of the published fit (3489 K), so no composition at this temperature lies inside a gap. The measurement therefore cannot detect that perturbation. It narrows the spinodal only through the functional forms of the four model classes. The curvature follows from the long-wavelength limit of the Bhatia–Thornton concentration–concentration structure factor, $S_{cc}(0) = RT/(\partial^2 g/\partial x^2)$ per mole of mixing units [1, 18]. Four practical points apply. Simulations count atoms, not mixing units, so the normalisation must be converted. The melt has several reacting species, so the full matrix of partial structure factors is needed. At fixed particle numbers, $S_{cc}(0)$ requires an extrapolation to zero wave number or a finite-volume Kirkwood–Buff correction [20]. And the cell must be large. Chen et al. state that structure-factor estimates of the curvature are nonnegative by construction and limited by system size and run

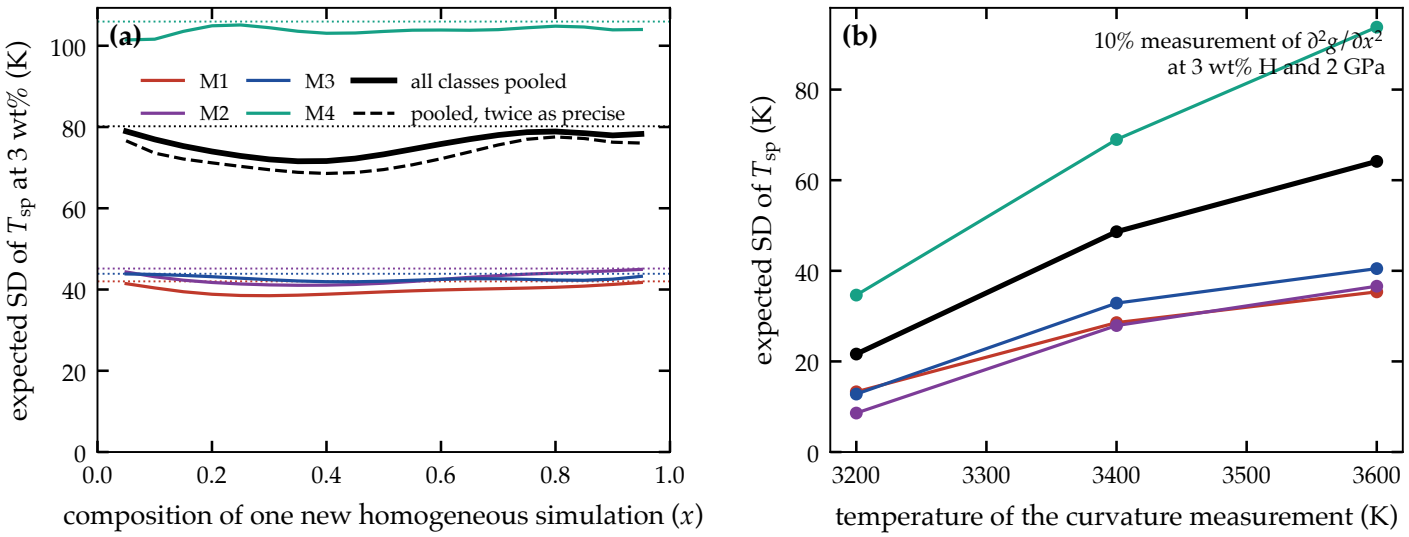


Figure 4.1: ***Simulations that would constrain the spinodal.*** *Expected posterior standard deviation of the spinodal temperature of a 3 wt% parcel at 2 GPa after (a) one new homogeneous simulation of excess enthalpy, entropy and volume at composition x with the precision of the ED Table 2 entries, for each model class fitted to all data (thin lines) and all classes pooled (thick line; dashed: pooled, twice that precision), with the current values as dotted lines, and (b) a measurement of the composition curvature $\partial^2 g/\partial x^2$ at the baseline composition with 10% precision, at three temperatures. Linear-Gaussian screen of the recalibration samples.*

time [8]. They do not run a size test. In H/He, $S(0)$ converges only in cells of about 1700 atoms or more, roughly 14 Å across ([37], Supplementary Information). Present DFT-MD cells are smaller, so a machine-learned potential trained on the same functional is the realistic route.

5. Conclusion

Hydrogen exsolution from a miscible sub-Neptune interior releases heat, and the evolution models we checked leave it out. We added energy, nucleation and cooling-path functions to Atmodeller and applied them to the 2026 H_2–$MgSiO_3$ mixing model and its source simulations. This chapter summarizes the findings and what they imply.

5.1 Summary of Findings

5.1.1 Open-Source Tools

The published model uses oxide units, whereas Atmodeller uses $MgSiO_3$ formula units. The conversion must act on the extensive free energy, not only on the compositions. With it, our implementation reproduces the published phase diagram with an RMS residual of 2.3 K. Without it, the metastable interval of the baseline parcel is about nine times too short. The new energy, nucleation and cooling-path functions are tested against independent solutions. The full Atmodeller equilibrium solver reproduces the mixing enthalpy along one published-fit parcel path to a relative difference of 9×10^{-16}.

5.1.2 The Heat of Exsolution

At 1–3 wt% hydrogen and 2–5 GPa, exsolution releases 35–93 MJ per kg of hydrogen in the class-M1 refits that use the enthalpy data, and 51–64 MJ kg^{-1} in the published fit. The simulated excess enthalpies anchor this value. They also show that the published fit puts too little excess enthalpy on the hydrogen-poor side and too much on the hydrogen-rich side. In a convecting interior with 3 wt% hydrogen, the refit M1 gives a heat over the first 300 K below the binodal of 2.3 times the sensible heat. Just below the binodal, exsolution adds up to 6.0 times the sensible heat capacity, so the effective heat capacity rises up to 7.0-fold if the hydrogen leaves as pure H_2. For a whole planet, the heat is 4–7% of the radiated energy in a published budget. This is a scaling estimate on a thermal history from an earlier phase diagram. A consistent planetary value needs an evolution model built on the 2026 free energy.

5.1.3 Exsolution Close to Equilibrium

The top of a convecting interior degasses into the envelope without nucleation. Away from the envelope, homogeneous nucleation ends metastability 1.2–2.9 K below the binodal for the published fit, and at most 1.3% of the hydrogen is withheld. Young et al. plotted the binodal–spinodal window for the 2025 fit [39]. In the 2026 fit it spans 414 K for the baseline parcel, but it is a mean-field upper bound that cooling parcels do not reach.

5.1.4 What the Source Data Constrain

The published fit departs from its source simulations in three places: the compositions of the two near-critical runs, the excess volume at 5 GPa and 4000 K, and the excess enthalpy of the most hydrogen-poor simulated mixture ($x = 0.24$) at the same conditions. Across four model classes and six data treatments, the median window ranges from 41 to 242 K. A perturbation that leaves every bulk equilibrium quantity unchanged still moves the spinodal by 4.9 to 124.4 K between 5 and 2 wt% hydrogen. At the 3 wt% baseline the shift is only 16.2 K, so there the spread between model classes dominates. The spinodal is therefore a property of the assumed functional form. The heat is better constrained, because it rests on the simulated enthalpies of homogeneous states.

5.2 Implications

1. **Energy budgets.** Evolution models with miscible interiors should include the heat of exsolution. It should come from the same free energy as the phase boundary, so that the two stay consistent.
2. **Timing of release.** Equilibrium exsolution at the binodal is a good approximation for convecting interiors. The demographic tests proposed for young sub-Neptunes [29] do not need the metastable window as a systematic uncertainty.
3. **Fitting mixing models.** A mixing model used for energies must be fitted to the excess enthalpies, not only to phase boundaries. A fit to phase compositions alone overpredicts the heat, and the enthalpy data exclude it.

5.3 Future Outlook

The per-run composition estimators, composition time series and interface widths of the source simulations would sharpen both the heat and the kinetics. Homogeneous enthalpies below 4000 K would test the temperature dependence of the heat. A machine-learned potential trained on the same functional could test the effect of cell size and map coexistence in large cells. It could also measure the composition curvature through the structure factor. The next step for the planet is an evolution model that uses the 2026 free energy for both the phase boundary and the energy budget. It would turn the interior result into a planetary cooling history. Helium, iron and the reactive species of the melt are the next components to add.

Appendices

A. Thermodynamics of the Mixing Model

This appendix presents a self-contained theoretical framework for the hydrogen–silicate mixing model used in `Atmodeller`. It unifies the free energy, the composition basis, the chemical potentials and the stability conditions. The goal is to provide a physically transparent explanation of why the binodal and the spinodal of the published model behave as they do, and of which parts of that behaviour come from the model form rather than from the data.

A.1 Chemical Potentials on the Oxide Basis

For a binary mixture of oxide units (component 1) and H_2 (component 2), with $g_{\mathrm{mix}}(x)$ per mole of mixing units and x the H_2 fraction, the chemical potentials relative to the pure components are

$$\mu_1 = g_{\mathrm{mix}} - x\, g'_{\mathrm{mix}}, \qquad \mu_2 = g_{\mathrm{mix}} + (1-x)\, g'_{\mathrm{mix}}. \tag{A.1}$$

For Eq. (2.4) with $f = 1 - T/\tau + P/\pi$, the excess parts are

$$\mu_1^{\mathrm{ex}} = f\, x^2\, [B + 2(A-B)(1-x)]\,, \tag{A.2}$$
$$\mu_2^{\mathrm{ex}} = f\, (1-x)^2\, [A + 2(B-A)x]\,, \tag{A.3}$$

which is equation 14 of [13] written for both components.

A.2 Conversion to Native Formula Units

`Atmodeller` counts $n_{\mathrm{MgSiO_3}}$ and $n_{\mathrm{H_2}}$. With $G = (2n_{\mathrm{MgSiO_3}} + n_{\mathrm{H_2}})\, g_{\mathrm{mix}}(x)$ and $x = n_{\mathrm{H_2}}/(2n_{\mathrm{MgSiO_3}} + n_{\mathrm{H_2}})$,

$$\frac{\partial G}{\partial n_{\mathrm{MgSiO_3}}} = 2\mu_1, \qquad \frac{\partial G}{\partial n_{\mathrm{H_2}}} = \mu_2. \tag{A.4}$$

Per native mole, $h(y) = (2-y)\, g_{\mathrm{mix}}(x(y))$ with $x = y/(2-y)$ and $dx/dy = 2/(2-y)^2$. Differentiating twice gives

$$h''(y) = \frac{4\, g''_{\mathrm{mix}}(x)}{(2-y)^3}. \tag{A.5}$$

The curvature changes magnitude but not sign, so the spinodal is the same in both bases, as it must be. The coexisting compositions are also the same once converted with Eq. (2.7). Importing the published parameters without the factor $2-y$, as in our incorrect-basis control, changes the free energy itself and therefore moves both curves.

A.3 Spinodal Temperature and Its Threshold

The excess curvature of Eq. (2.5) is linear in composition,

$$W''(x) = 2B - 4A + 6(A-B)\, x. \tag{A.6}$$

Setting $g''_{\mathrm{mix}} = 0$ gives Eq. (2.8). A positive spinodal temperature requires $W''(x) < 0$, that is

$$x > x_{\min} = \frac{2B - 4A}{6(B-A)} = 0.262 \tag{A.7}$$

for the published parameters. This corresponds to $w = 1.40$ wt% local hydrogen. Below this threshold a parcel can still separate in equilibrium, but its homogeneous state never becomes unstable: separation there always requires nucleation.

A.4 Why the Withheld Fraction Does Not Depend on Pressure

Write $g_{\mathrm{mix}}/(RT) = s(x) + r\,W(x)$, with s the ideal term and

$$r(T,P) = \frac{1 - T/\tau + P/\pi}{RT}. \tag{A.8}$$

Phase equilibrium and stability depend on T and P only through r. At fixed composition the spinodal fixes $r_{\mathrm{sp}} = -1/[x(1-x)W''(x)]$, and the binodal fixes another value $r_b(x)$. Solving $r(T,P) = r_*$ gives

$$T_*(x,P) = \frac{1 + P/\pi}{R\,r_* + 1/\tau} = T_*(x,0)\left(1 + \frac{P}{\pi}\right). \tag{A.9}$$

Both onset temperatures therefore scale with $1 + P/\pi$, and the coexisting compositions at the spinodal, hence f_{H}, are the same at every pressure. The numerical invariance of f_{H} (Section 3.2) is a consequence of this separable form, not independent evidence. Classes M3 and M4 share it, because all their terms scale with the same factor; class M2 does not.

A.5 Critical Point

At the critical point both the second and third derivatives of g_{mix} vanish. For Eq. (2.4) this gives equations 17–19 of [13]: the ratio B/A fixes the critical composition, here $x_C = 0.720$ ($w = 9.36\,\mathrm{wt\%}$), and

$$T_C(P) = T_C^0\left(1 + \frac{P}{\pi}\right), \qquad T_C^0 = 3700\,\mathrm{K}. \tag{A.10}$$

The critical temperature is the same whichever component carries A; the critical composition is not.

A.6 Curvature Change of the Identifiability Perturbation

For the perturbation of Eq. (2.13) with gap width $\Delta x = x_{\mathrm{rich}} - x_{\mathrm{poor}}$,

$$\frac{\partial^2 \delta g}{\partial x^2} = \lambda B_{\mathrm{gap}}\,\frac{384\,u(1-u)(1-5u+5u^2)}{\Delta x^2}. \tag{A.11}$$

The change is negative in the middle of the gap ($0.28 < u < 0.72$) and positive near its edges, so the perturbation can move the spinodal in either direction. This is why the 3 wt% spinodal rises while the 5 wt% spinodal falls in Figure 3.6. The value, first and second derivatives of δg vanish at $u = 0$ and $u = 1$, which keeps every equilibrium quantity unchanged.

A.7 Energies of Mixing and Separation

Atmodeller obtains the mixing enthalpy, entropy and volume from the native Gibbs energy by automatic differentiation, $h = g - T\,\partial g/\partial T$, $s = -\partial g/\partial T$ and $v = \partial g/\partial P$. For the published form these reduce to $h = W(x)(1 + P/\pi)$ and $v = W(x)/\pi$ per oxide-basis mole, and the mixing heat capacity vanishes. The partial molar enthalpy of component i is $\bar{h}_i = -RT^2\,\partial \ln \gamma_i/\partial T$. For H_2 on the oxide basis,

$$\bar{h}_{\mathrm{H_2}} = (1-x)^2\,[A + 2(B-A)\,x]\,(1 + P/\pi). \tag{A.12}$$

The equilibrium enthalpy of a two-phase state is the lever-rule average of the two phase enthalpies. Its temperature derivative at fixed bulk composition is the apparent heat capacity, which includes the heat released as the phase compositions move. By the envelope theorem the lever-rule entropy and volume equal the temperature and pressure derivatives of the equilibrium Gibbs energy; the tests check this to 10^{-9}. The separation heat of a homogeneous parcel, $h_{\mathrm{hom}} - h_{\mathrm{eq}}$, is zero outside the gap and continuous at the binodal.

For the convecting interior, the melt composition follows the hydrogen-poor binodal at the top pressure and the exsolved hydrogen ends as pure H_2. The heat released between two top temperatures is then the change of the melt's total mixing enthalpy, which equals $\int \bar{h}_{\mathrm{H_2}}\,\mathrm{d}n_{\mathrm{H_2}}$ because the mixing heat capacity is zero.

A.8 Square-Gradient Nucleation

The free energy of a nonuniform parcel is $F = v^{-1}\int [\Delta g(x) + \kappa\,|\nabla x|^2]\,\mathrm{d}V$, where $\Delta g(x) = g(x) - g(x_0) - g'(x_0)(x - x_0)$ is measured from the tangent at the parent composition x_0 and v is the volume of one oxide-basis mole [6, 7]. In units of $\ell = (\kappa/RT)^{1/2}$ and RT, the critical nucleus solves the radial Euler–Lagrange equation $x'' + 2x'/r = \tfrac{1}{2}\,\partial(\Delta g/RT)/\partial x$, with $x'(0) = 0$ and $x \to x_0$ far away. We find it by shooting on the central composition with bisection between overshoot and undershoot. The centre can lie exponentially close to the nucleus composition, so the first part of the profile uses the linearised solution $\propto \sinh(mr)/(mr)$. The barrier follows from the virial identity, $W^* = \tfrac{2}{3}\,v^{-1}\int \kappa\,|\nabla x|^2\,\mathrm{d}V$, and the direct integral checks it to 2%. Because lengths scale with ℓ, $W^* \propto \kappa^{3/2}/v$ exactly, so one dimensionless table per parameter sample serves every κ and v. The classical estimate uses the planar tension $\sigma = 2\,v^{-1}(\kappa RT)^{1/2}\int (\Delta g_c/RT)^{1/2}\mathrm{d}x$ between the coexisting phases. A 10–90% width w of the planar profile gives $\kappa = RT\,(w/I)^2$, with $I = \int_{x_{10}}^{x_{90}} (\Delta g_c/RT)^{-1/2}\mathrm{d}x$. The solver agrees with an independent finite-volume Newton solution to 10^{-3}, tends to the classical barrier near the binodal, and vanishes as the 3/2 power at the spinodal.

B. Source Data and Audit Tables

This appendix reproduces the source simulations of Gilmore & Stixrude [13] that enter the audit and the recalibration, next to the predictions of the published fit. The phase-coexistence runs contain 27 $MgSiO_3$ formula units and 92 to 219 hydrogen atoms; the homogeneous runs contain 135 to 160 atoms. All compositions are H_2 mole fractions x on the oxide basis (Eq. 2.6).

B.1 Phase-Coexistence Runs

Table B.1 lists all 14 phase-coexistence runs. For runs that kept two phases it gives the composition of the H-poor and H-rich phase from ED Table 1 and, where they differ, from the digitized Figure 2a. For runs that became homogeneous it gives the bulk composition in parentheses. The last column is the published critical temperature at the run pressure minus the run temperature: positive values mean the model places the run below T_C.

Table B.1: ***Phase-coexistence runs of ED Table 1 and the published fit.*** *Pressure uncertainties in the last digits are given in parentheses. "same" means that the digitized Figure 2a agrees with the table to within 0.001.*

P (GPa)	T (K)	Phases	ED Table 1 x	Figure 2a x	Published fit x	$T_C - T$ (K)
2.05(4)	3000	two	0.209, 0.931	same	0.231, 0.990	+484
1.65(24)	3500	two	0.458, 0.576	0.419, 0.629	0.595, 0.828	+26
1.89(22)	4000	one	(0.522)	–	one phase	−500
1.79(12)	5000	one	(0.460)	–	one phase	−1489
1.48(7)	6000	one	(0.471)	–	one phase	−2456
1.41(3)	8000	one	(0.565)	–	one phase	−4449
5.14(10)	3000	two	0.587, 0.706	0.514, 0.736	0.398, 0.941	+157
4.71(5)	3500	one	(0.616)	–	one phase	−298
4.06(5)	4000	one	(0.597)	–	one phase	−729
3.95(5)	5000	one	(0.583)	–	one phase	−1717
9.58(10)	2500	two	0.316, 0.976	same	0.346, 0.961	+187
10.42(6)	3000	one	(0.670)	–	one phase	−401
9.29(14)	4000	one	(0.660)	–	one phase	−1282
7.85(28)	6000	one	(0.638)	–	one phase	−3130

In every two-phase run the per-phase atom numbers of ED Table 1 add up to the full simulation cell, to within one atom. Each atom, including those in the two interfaces, is therefore assigned to one of the two phases. For the 2.05 and 9.58 GPa runs the table and the figure agree to within 0.0006. For the 1.65 and 5.14 GPa runs, which lie closest to the critical curve, they differ by up to 0.07.

B.2 Homogeneous Excess Properties

Table B.2 lists the excess properties reconstructed from ED Table 2 per mole of mixing units. The values include the pressure-offset correction described in Section 2.5, and the uncertainties include the shared endmember errors. The enthalpy entries of different compositions are correlated with coefficients of 0.45 to 0.75, the volumes with 0.44 to 0.78.

Table B.2: ***Excess properties of the homogeneous simulations at 5.06–5.08 GPa and 4,000 K*** *(ED Table 2), with 1σ uncertainties, next to the published fit.*

	H_{ex} (kJ mol^{-1})		S_{ex} (J mol^{-1} K^{-1})		V_{ex} (cm^3 mol^{-1})	
x	simulation	fit	simulation	fit	simulation	fit
0.833	36.6 ± 3.6	43.5	10.6 ± 3.1	11.5	-0.31 ± 0.12	-1.45
0.588	61.8 ± 4.2	58.0	13.9 ± 4.9	15.4	-0.94 ± 0.16	-1.94
0.238	36.3 ± 5.9	24.5	7.6 ± 6.1	6.5	-0.39 ± 0.26	-0.82

C. Recalibration Details

This appendix documents the priors, the sampler and the best-fitting parameters of the recalibration of Section 2.5.

C.1 Priors

All priors are uniform. For class M1, A lies between -300 and 1,500 kJ mol^{-1} and B between -300 and 3,000 kJ mol^{-1}. For the other classes the leading coefficients lie between -300 and 3,000 kJ mol^{-1}, and higher Redlich–Kister terms within $\pm$2,000 kJ mol^{-1}. The inverse temperature scale $1/\tau$ lies between 0 and 10^{-3} K^{-1} and the inverse pressure scale $1/\pi$ between -0.25 and 0.1 GPa^{-1}. For class M2 the subregular entropy coefficients lie between -200 and 800 J mol^{-1} K^{-1} and the volume coefficients between -60 and 30 cm^3 mol^{-1}. Parameter sets for which the homogeneous free energy develops more than one unstable composition interval at any evaluated state are outside every model class and receive zero probability. The evaluated states are the 14 runs and the spinodal of the baseline parcel, where the lever rule gives f_H.

C.2 Sampler and Convergence

Each fit starts from the best of eight Nelder–Mead optimizations around the published parameters. Eight adaptive Metropolis chains [16] then run 12,000 steps each, each from the best fit plus a Gaussian offset with a standard deviation of 2% of each prior range. During the first 3,000 steps the proposal covariance and scale adapt; these steps are discarded. Every 30th later step is kept, which gives 2,400 posterior samples per fit. The acceptance rate is about 0.2 in every fit. The largest classic split-chain $\hat{R}$ [12] over all parameters and derived quantities of all fits is 1.016, for B in the M1 fit to phase states and compositions; $1/\tau$ in that fit also exceeds the threshold of 1.01 of Section 2.5. Of the nine fits, 1 lies above this threshold (Section 3.4).

C.3 Best-Fitting Parameters

Table C.1 lists the best-fitting (MAP) parameters of class M1 for each data treatment, next to the published values.

D. Reproducibility

This appendix lists how every number and figure in this draft is produced. All paths are relative to the repository root. The analysis used Python 3.13 with the package versions recorded in `research/spinodal_onset/outputs/provenance.json`.

Table C.1: ***Best-fitting parameters of the published model form (class M1).*** *B/A sets the asymmetry and the critical composition; τ and π set the temperature and pressure scales.*

Fit	A (kJ mol^{-1})	B (kJ mol^{-1})	B/A	τ (K)	π (GPa)
Published fit	75	424	5.65	4400	−35
M1, all data	196	297	1.52	4252	−49
M1, Fig. 2a compositions	193	296	1.54	4285	−49
M1, $\sigma_x = 0.15$ near T_C	132	330	2.49	4429	−60
M1, without volumes	231	350	1.52	4274	−37
M1, phases + compositions	221	1403	6.36	3684	−35
M1, phases + homogeneous	227	319	1.41	4495	−69

D.1 Pipeline

1. `run_study.py` computes the onset grid, the cooling paths, the source-data audit, the homogeneous audit, the stress cases and the native `Atmodeller` solves.
2. `identifiability.py` builds the perturbation of Eq. (2.13) and checks that it leaves every equilibrium quantity unchanged.
3. `digitize_source_figure.py` digitizes Figure 2a of [13] and compares it with our implementation.
4. `calibration.py` runs the recalibration (Section 2.5). The environment variable `SPINODAL_WORKERS` sets the number of worker processes (default 2); with eight it takes about 40 minutes.
5. `design.py` computes the design screen of Section 4.5.
6. `make_figures.py` writes the study figures and `outputs/key_results.json`.
7. `make_paper_figures.py` writes all figures, and `paper/make_numbers.py` writes the computed numbers and table bodies to `paper/numbers.tex` and `paper/tables`.
8. `heat_study.py` computes the heat, nucleation and cooling results of Sections 3.6–3.9 in stages: `energy`, `edtable2`, `interior`, `barriers`, `delays`, `paths`, `checks` and `planet`. The `checks` stage holds the sensitivity checks of Section 3.9 and the interface width for half the window; the `planet` stage integrates the heat along the digitized evolution of [30]. `heat_figures.py` draws Figures 3.7, 3.8 and 3.9, and `paper/make_heat_numbers.py` writes `paper/heat_numbers.tex`.

Most computed numbers in the text are macros generated from the output files, so rebuilding the draft after a rerun updates the text with the figures. Some values are typed directly. Source values, such as the bulk compositions of source runs, are listed in Appendix B, and values from other studies are cited where they appear. Rounded ratios follow from the macros. The remaining typed values are one-off checks and estimates that no script in the pipeline regenerates:

- the pressure-offset and likelihood checks of Section 2.5 and the $\hat{R}$ of $1/\tau$ in Section 3.4;
- the gradient-transfer, cell-size and excess-volume estimates of Chapter 4.

The abstract and the conclusion repeat some of them. They must be rechecked by hand after each rerun.

D.2 Commands

```
uv sync --locked --all-extras --dev
PYTHONPATH=. uv run python research/spinodal_onset/run_study.py
PYTHONPATH=. uv run python research/spinodal_onset/identifiability.py
PYTHONPATH=. uv run python research/spinodal_onset/digitize_source_figure.py
SPINODAL_WORKERS=2 PYTHONPATH=. uv run python research/spinodal_onset/calibration.py
PYTHONPATH=. uv run python research/spinodal_onset/design.py
PYTHONPATH=. uv run python research/spinodal_onset/make_figures.py
PYTHONPATH=. uv run python research/spinodal_onset/make_paper_figures.py
PYTHONPATH=. uv run python research/spinodal_onset/paper/make_numbers.py
for s in energy edtable2 interior; do
  PYTHONPATH=. uv run python research/spinodal_onset/heat_study.py $s; done
PYTHONPATH=. uv run python research/spinodal_onset/heat_study.py barriers --start 0 --stop 201
```

```
PYTHONPATH=. uv run python research/spinodal_onset/heat_study.py barriers --start 201 --stop 402
for s in delays paths checks planet; do
  PYTHONPATH=. uv run python research/spinodal_onset/heat_study.py $s; done
cd research/spinodal_onset && PYTHONPATH=../.. uv run python heat_figures.py && cd ../..
PYTHONPATH=. uv run python research/spinodal_onset/paper/make_heat_numbers.py
cd research/spinodal_onset/paper && tectonic -X compile main.tex
```

The energy, nucleation and cooling functions are tested in `tests/test_cooling.py`, `tests/thermodata/test_mixing_energy.py` and `tests/thermodata/test_nucleation.py`.

D.3 Data Provenance

The ED Tables and Figure 2 of [13] are stored unmodified in `research/spinodal_onset/sources` with their SHA-256 hashes, source URLs and the CC-BY-4.0 attribution. No source value was changed. Where the table and the figure disagree, both are kept and analysed as separate data versions. The fiducial evolution and energy budget of [30] (their Figures 5 and 6) are stored as digitized curves, with the URLs and SHA-256 hashes of the source figures; their digitization error is a few per cent. The thesis layout reuses the KU Leuven title-page logo of the bachelor thesis. Its use should be checked against the current KU Leuven style rules before the draft is submitted or shared beyond this repository.

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